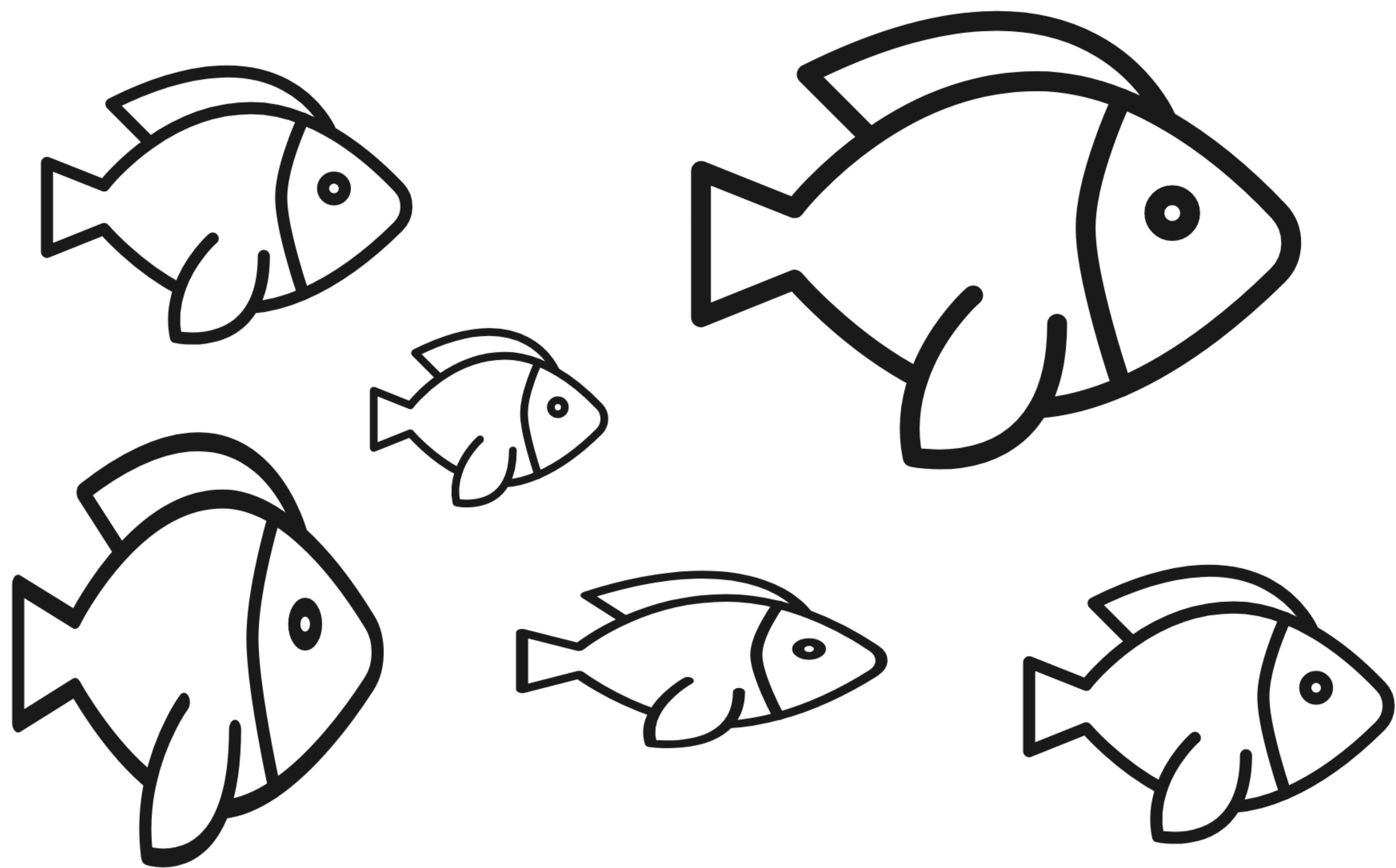
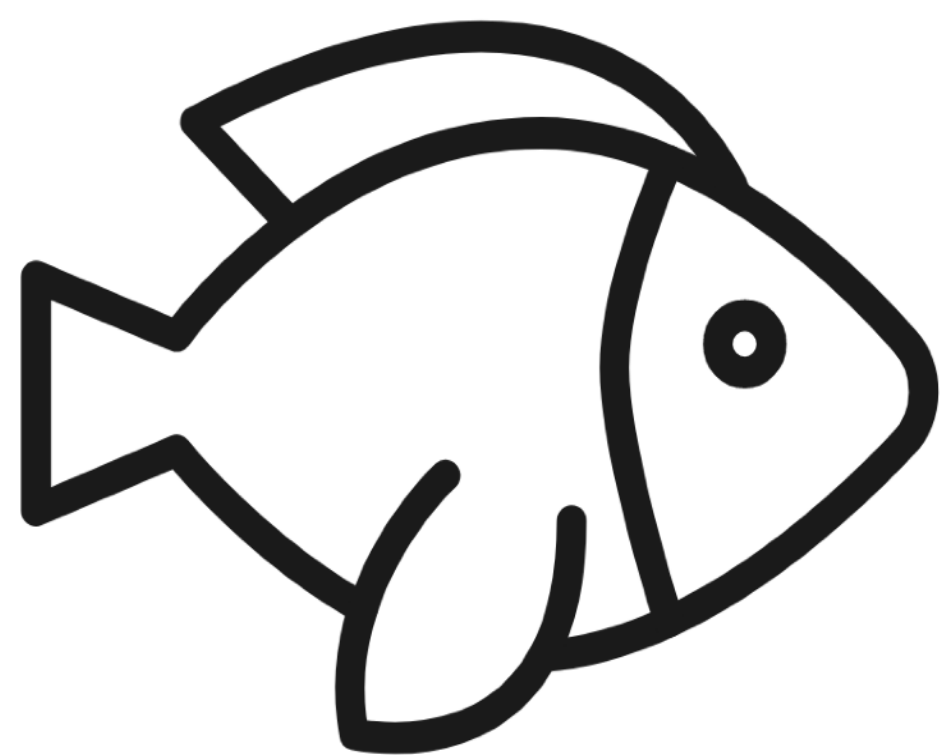
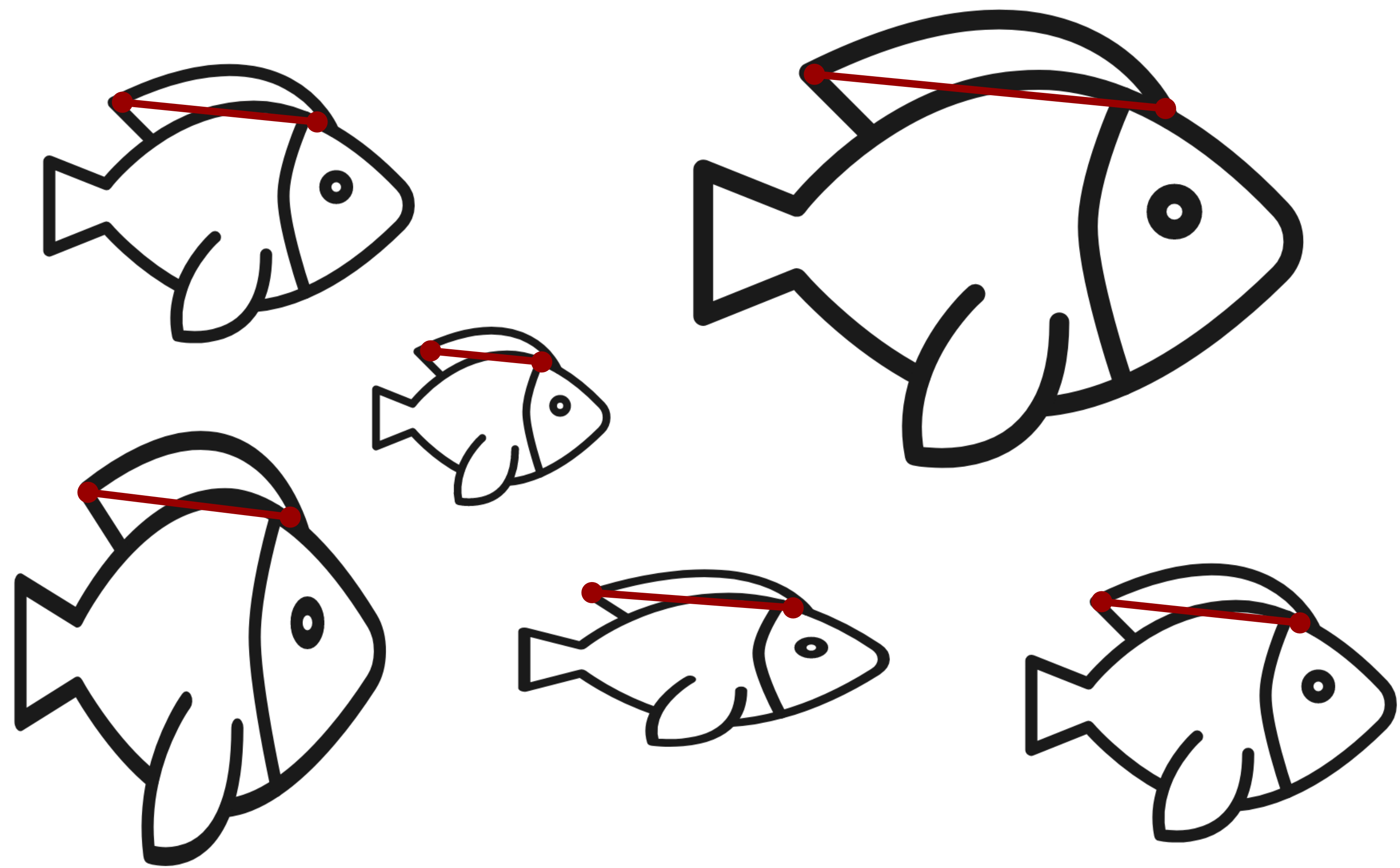


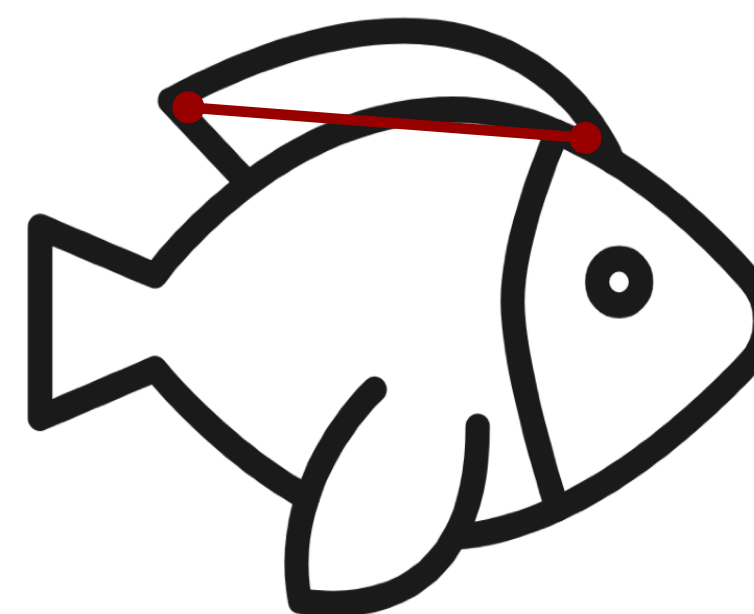
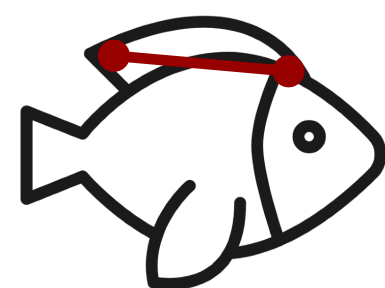
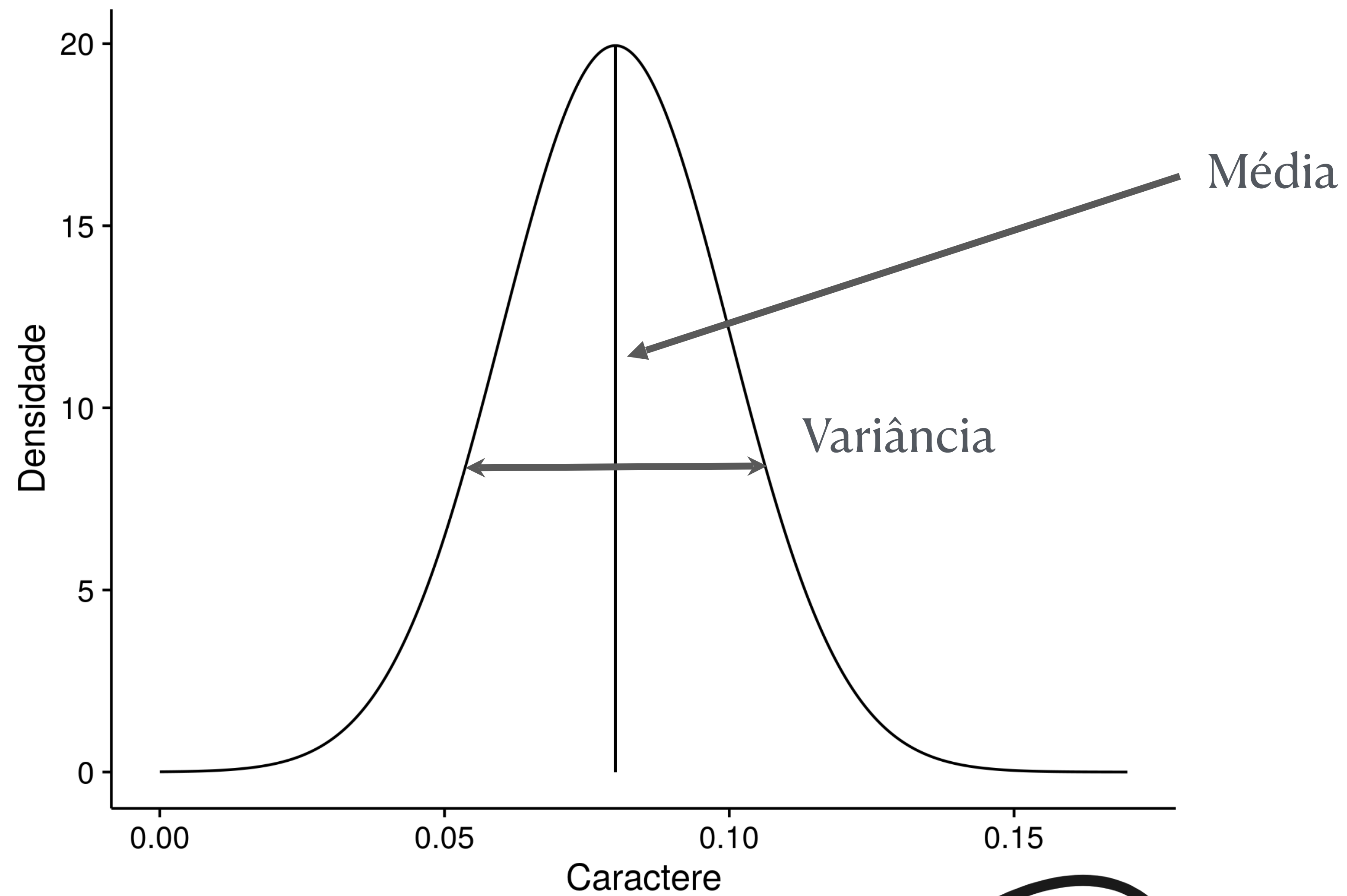
Quantificando variação multivariada

Variação e covariação

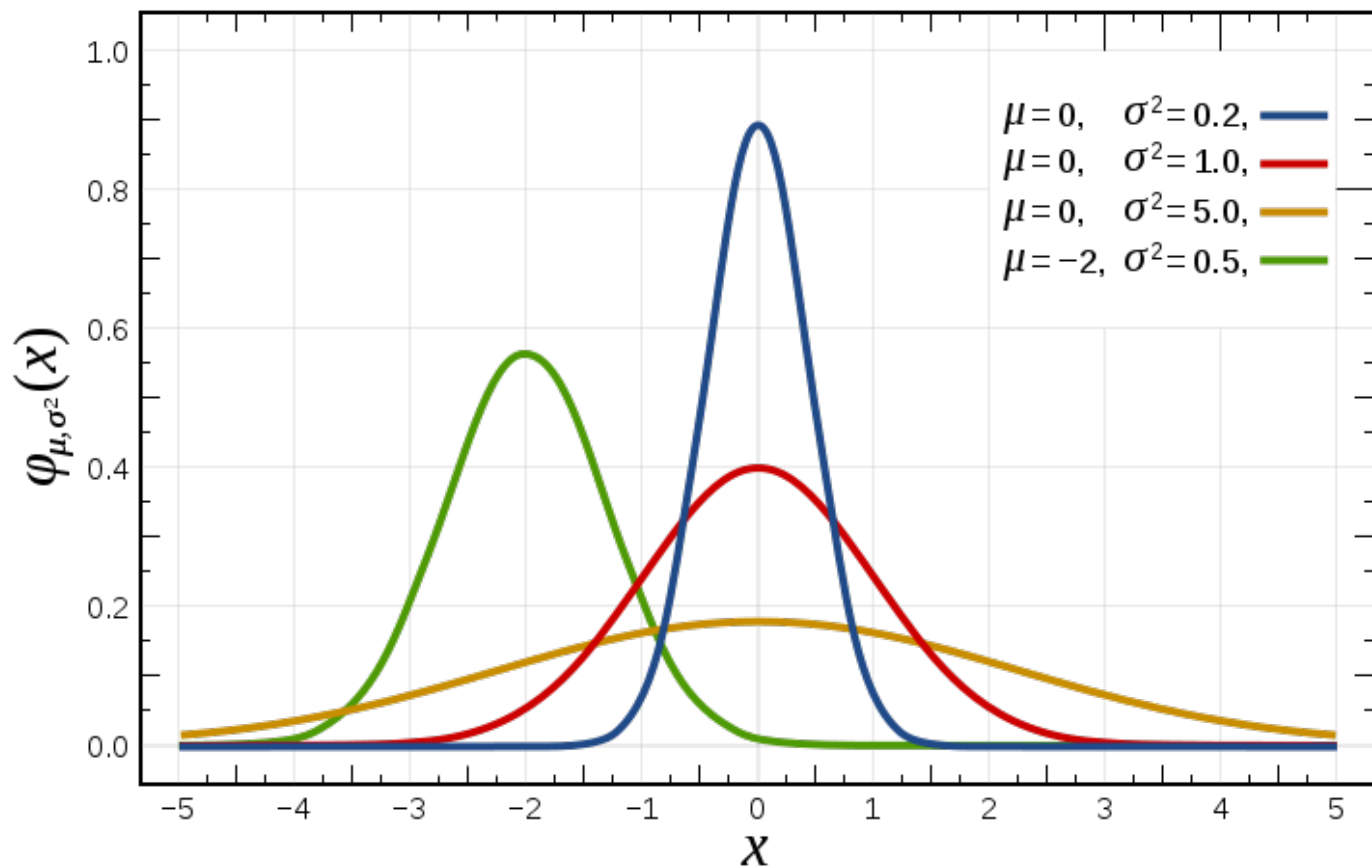






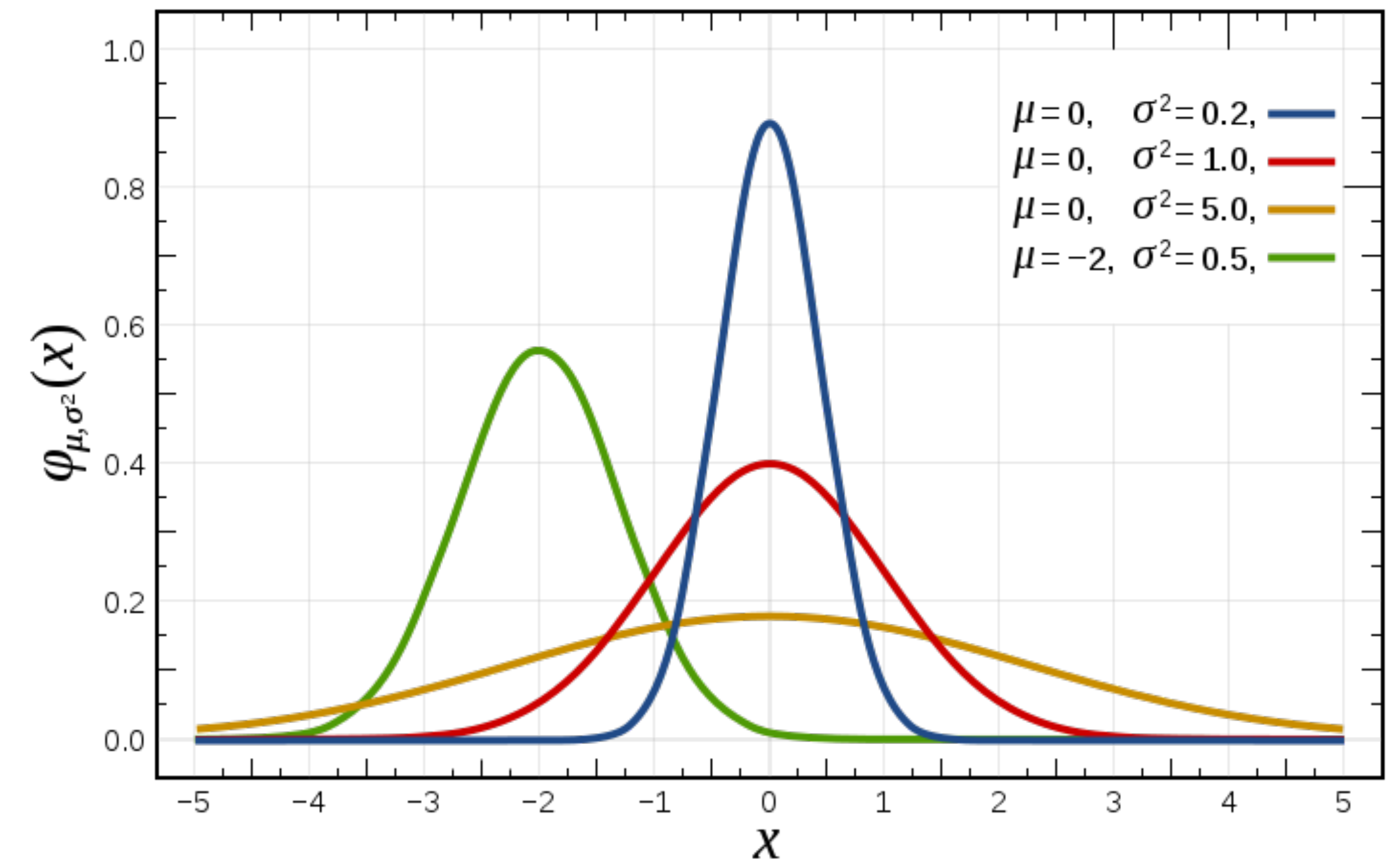


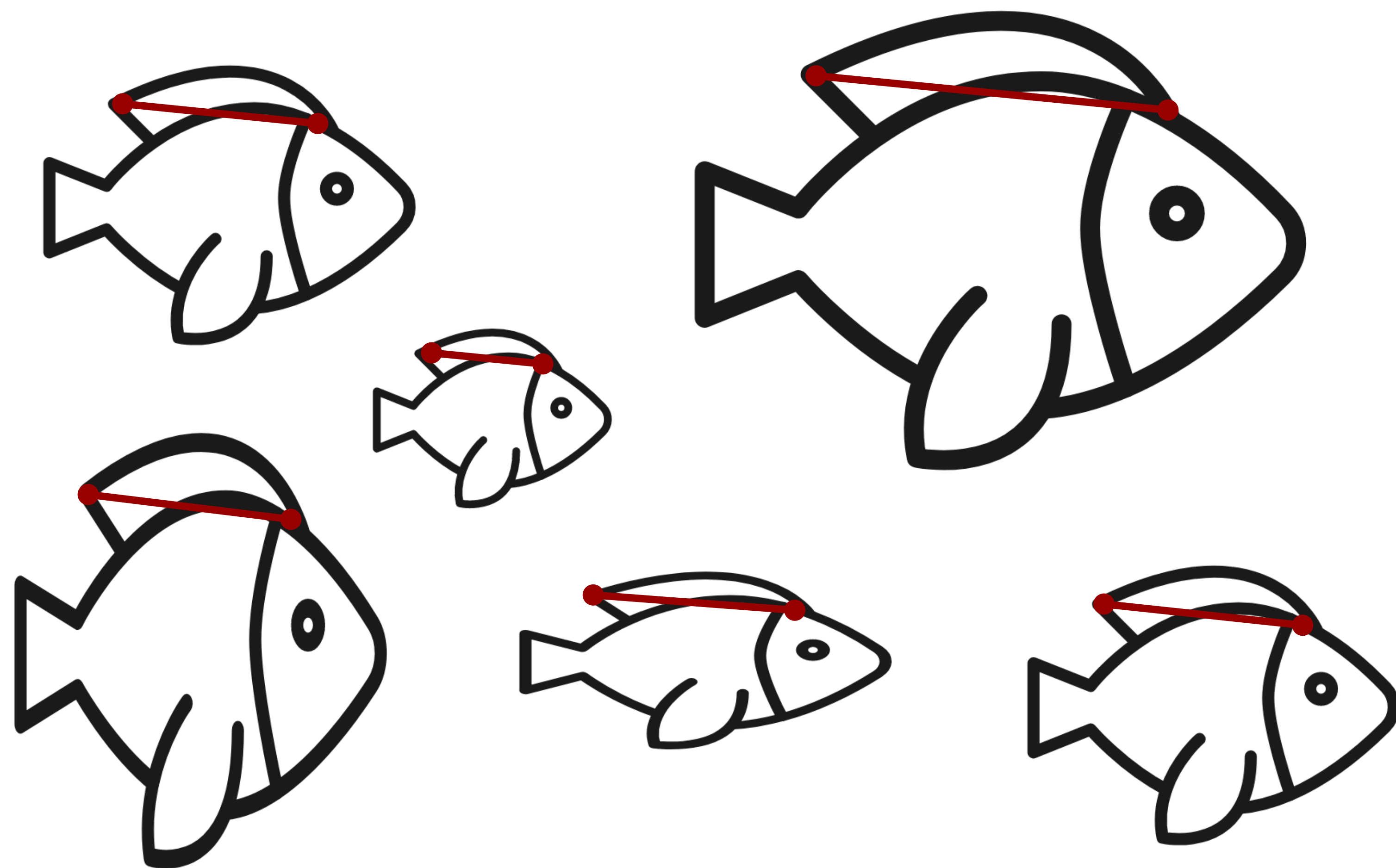
$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

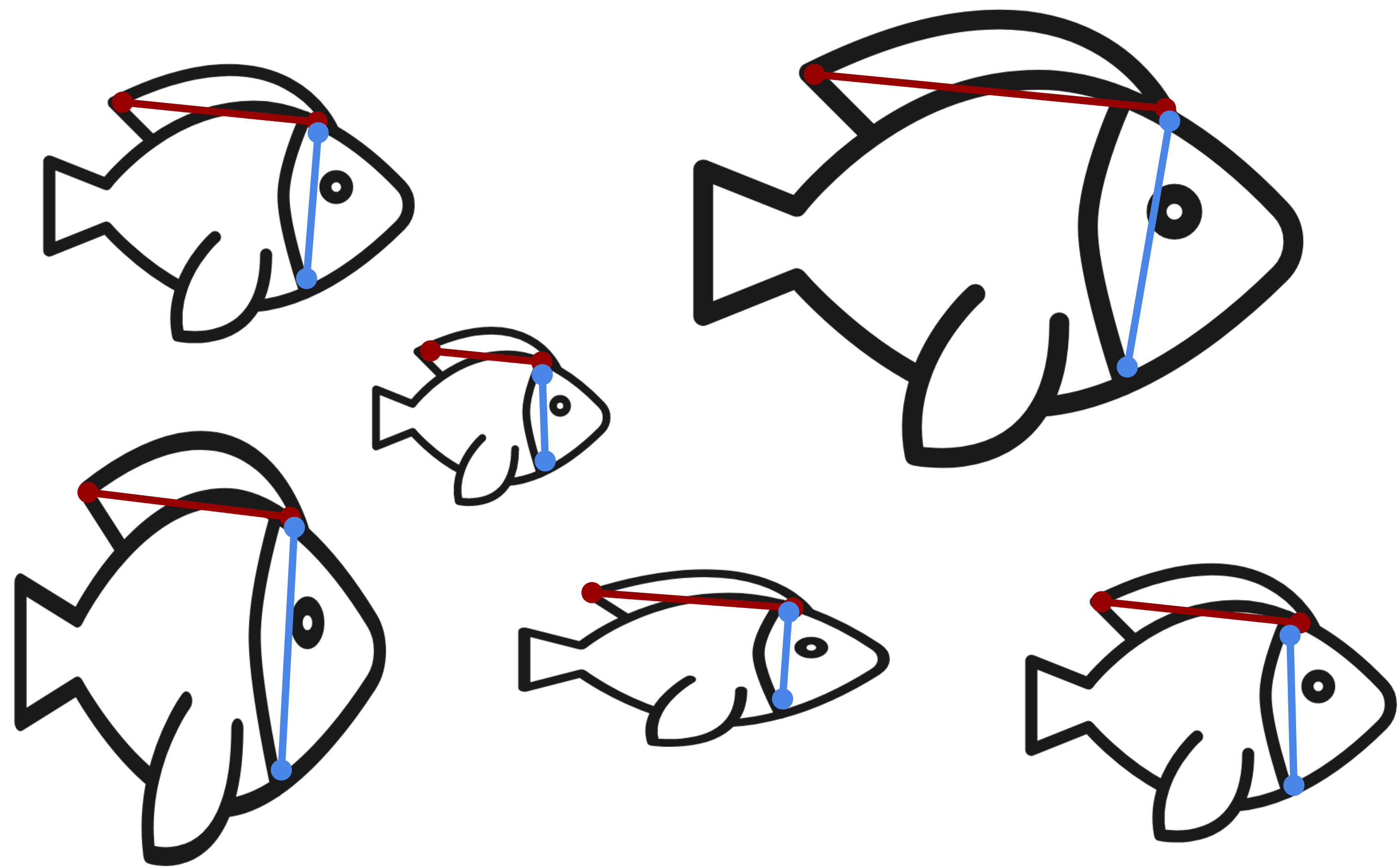


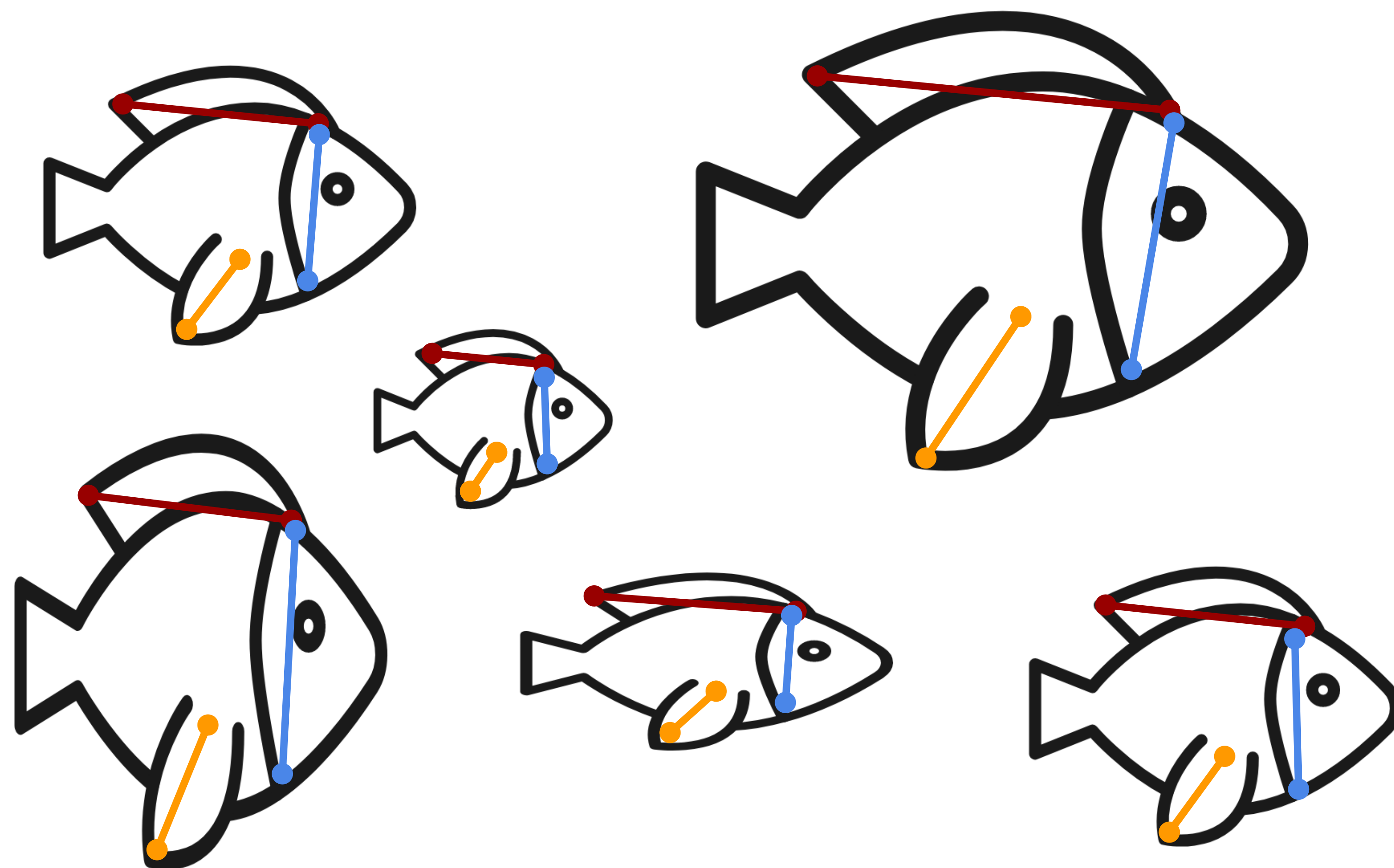
Em uma dimensão...

- Média e variância descrevem completamente o problema (em dados normais)
- Variância pode ser particionada em diferentes componentes
- Média e variância podem ser usados para comparar populações

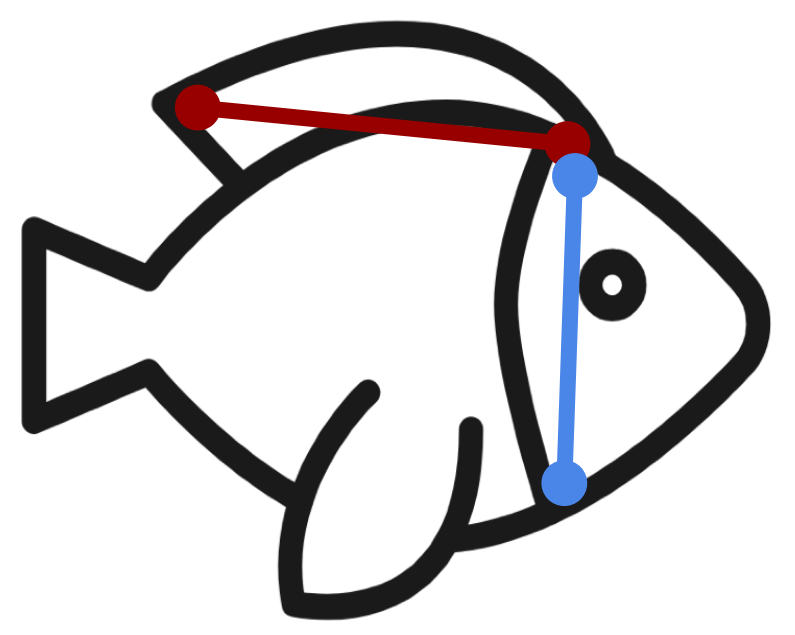
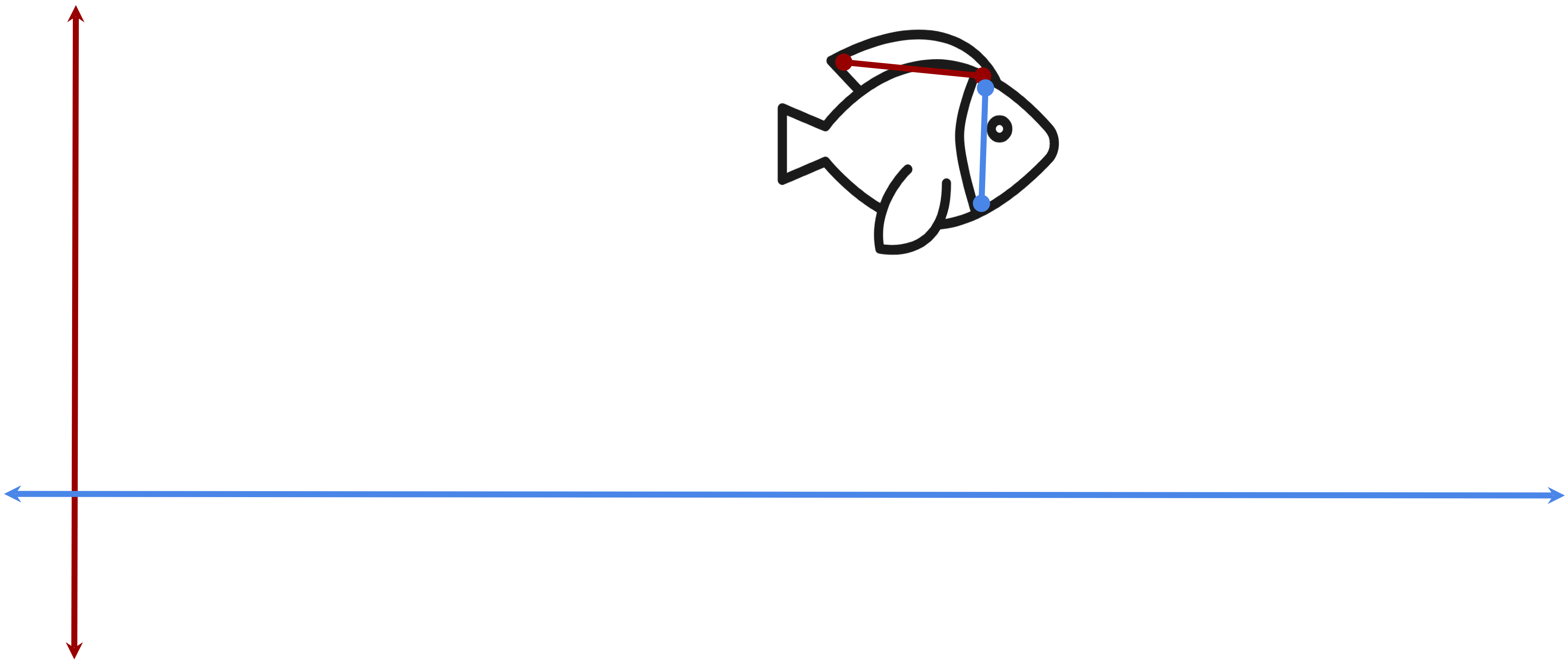


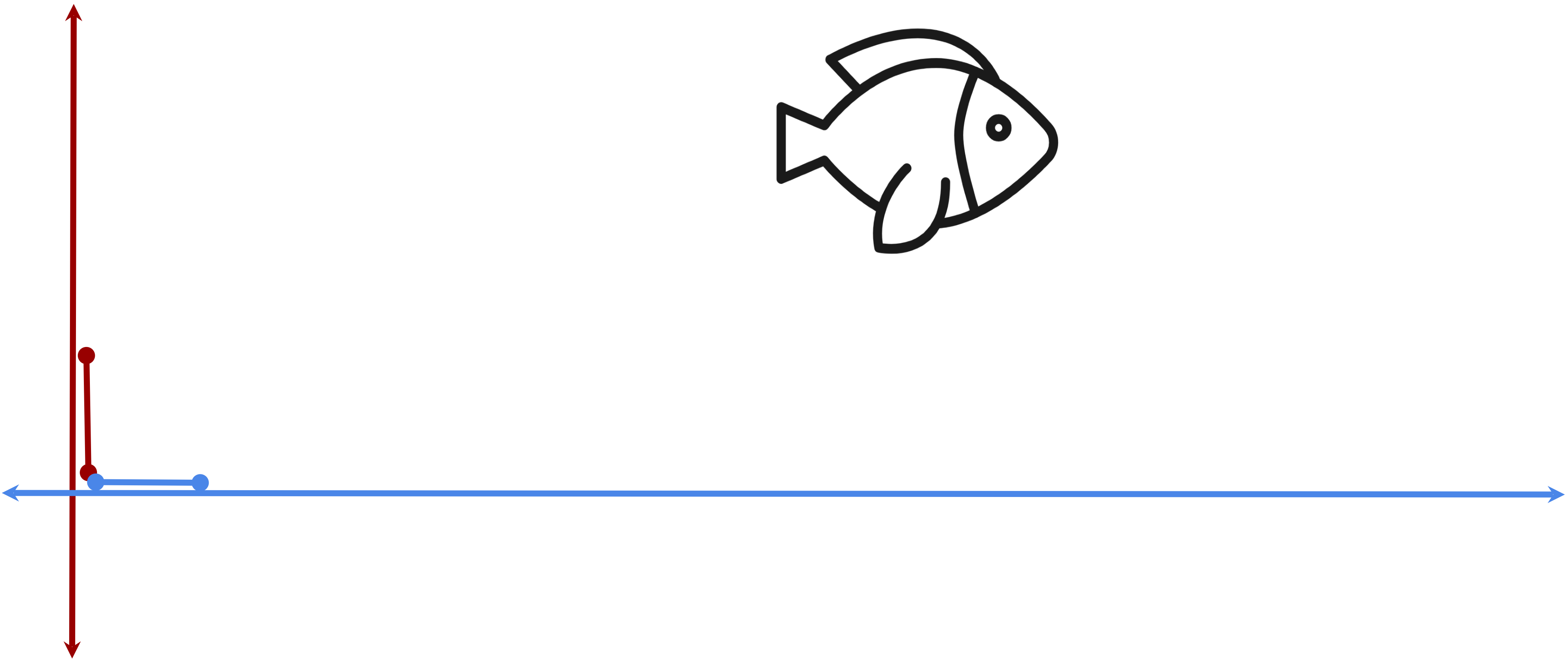
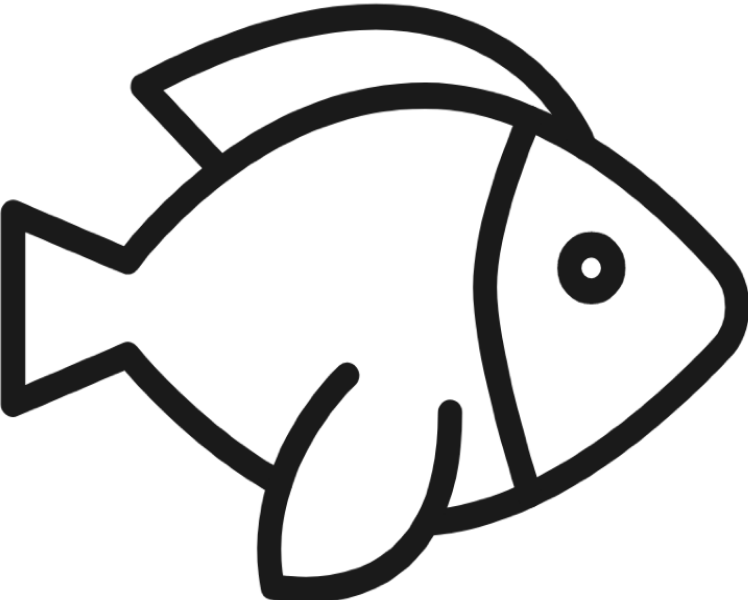


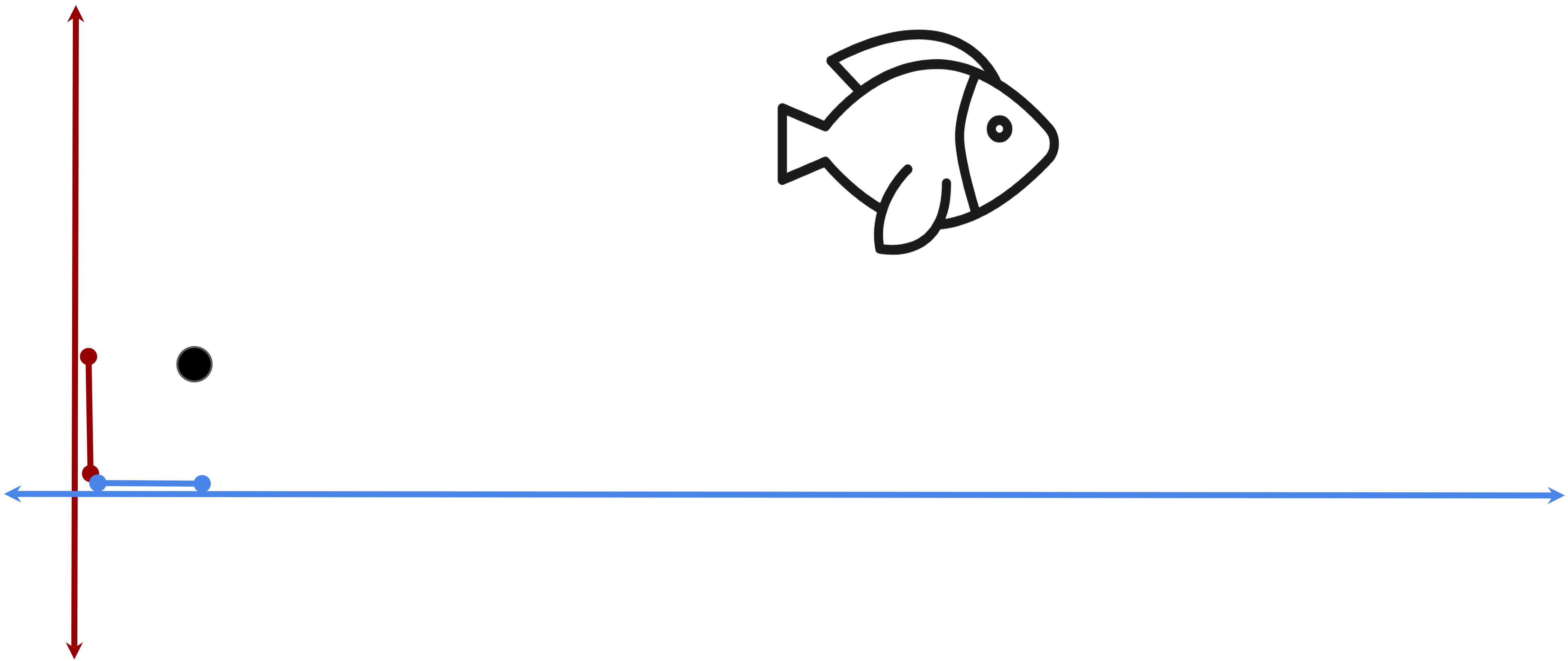




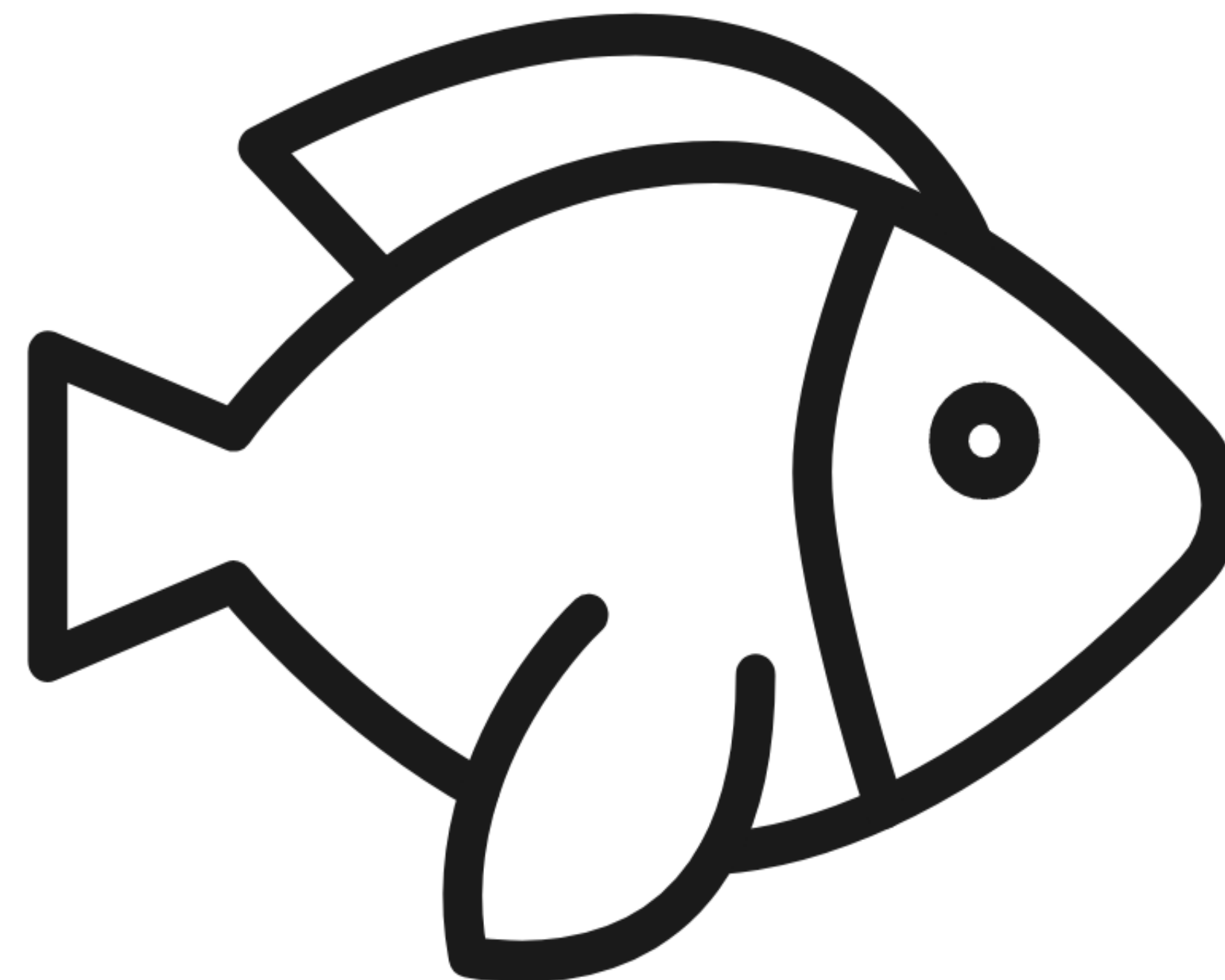
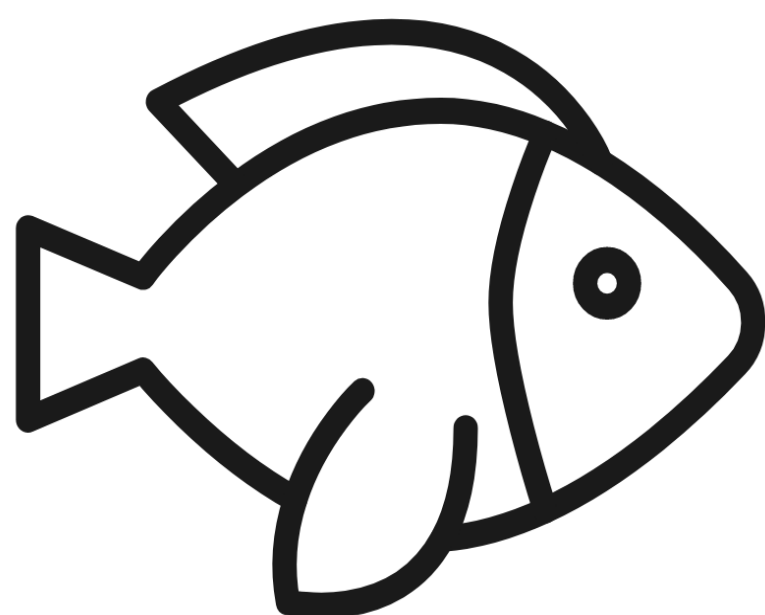


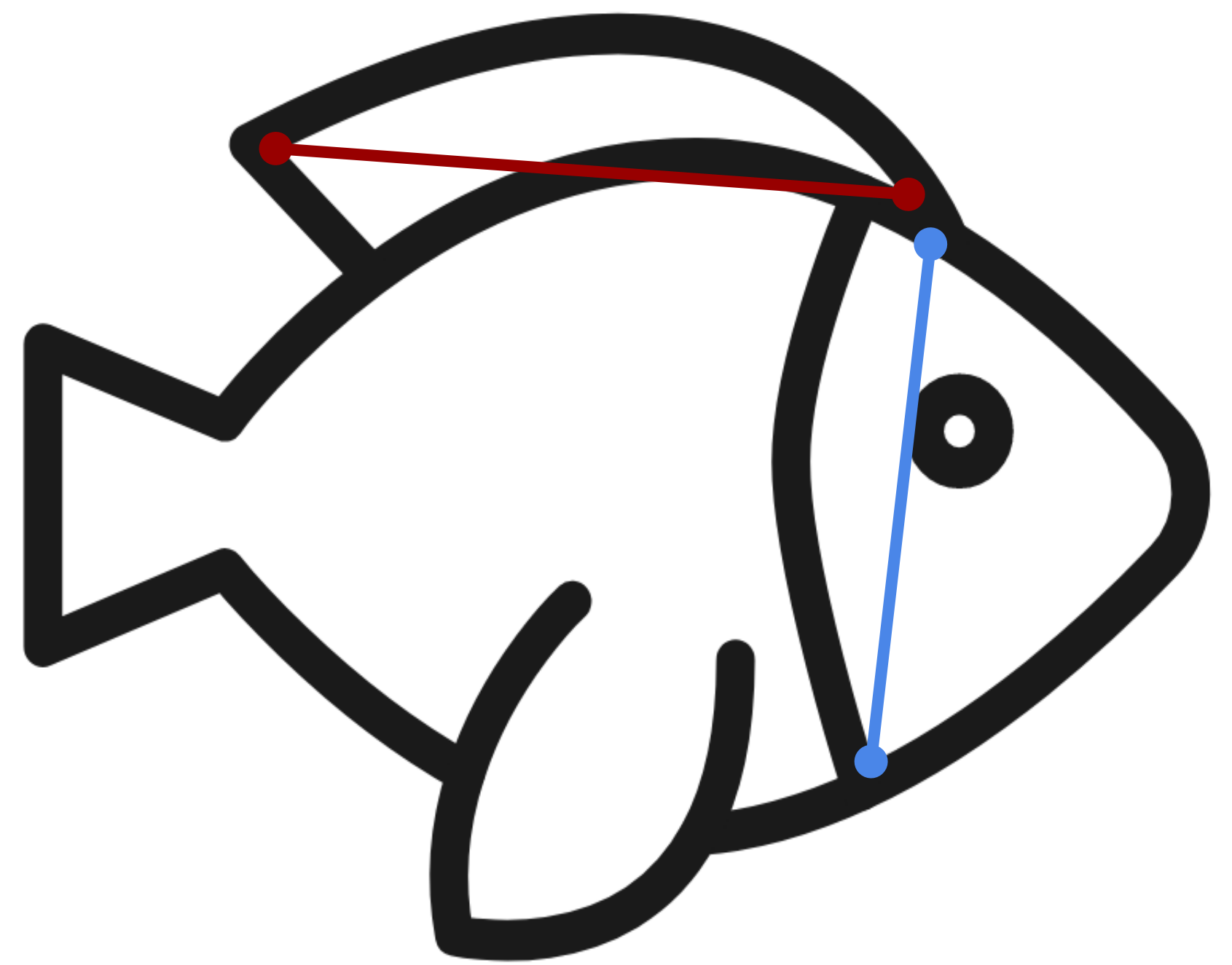
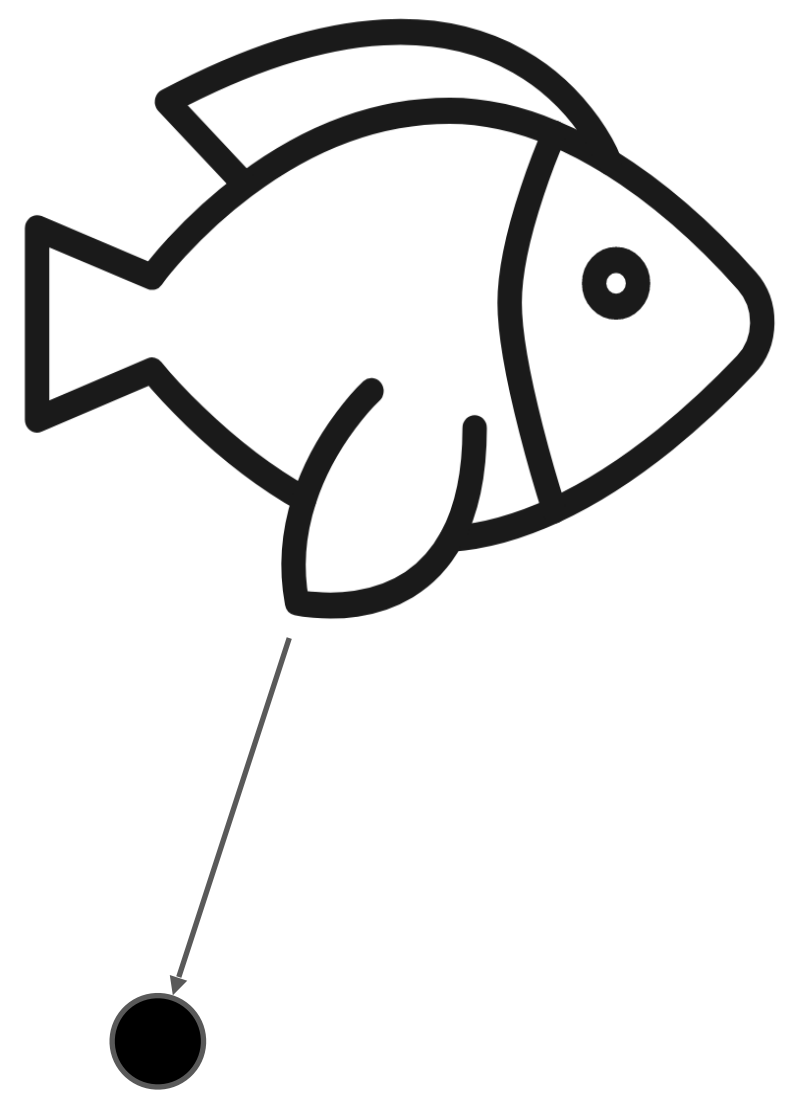


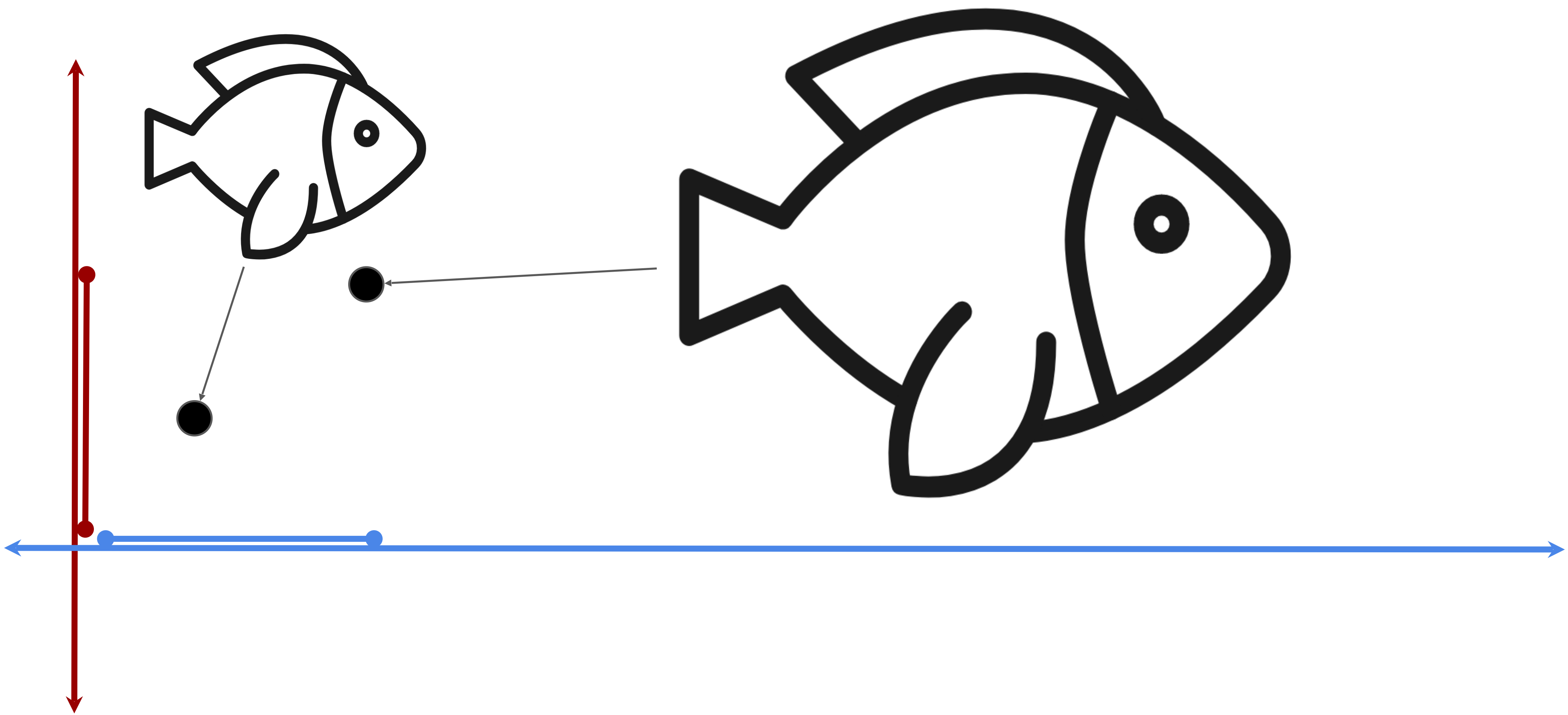






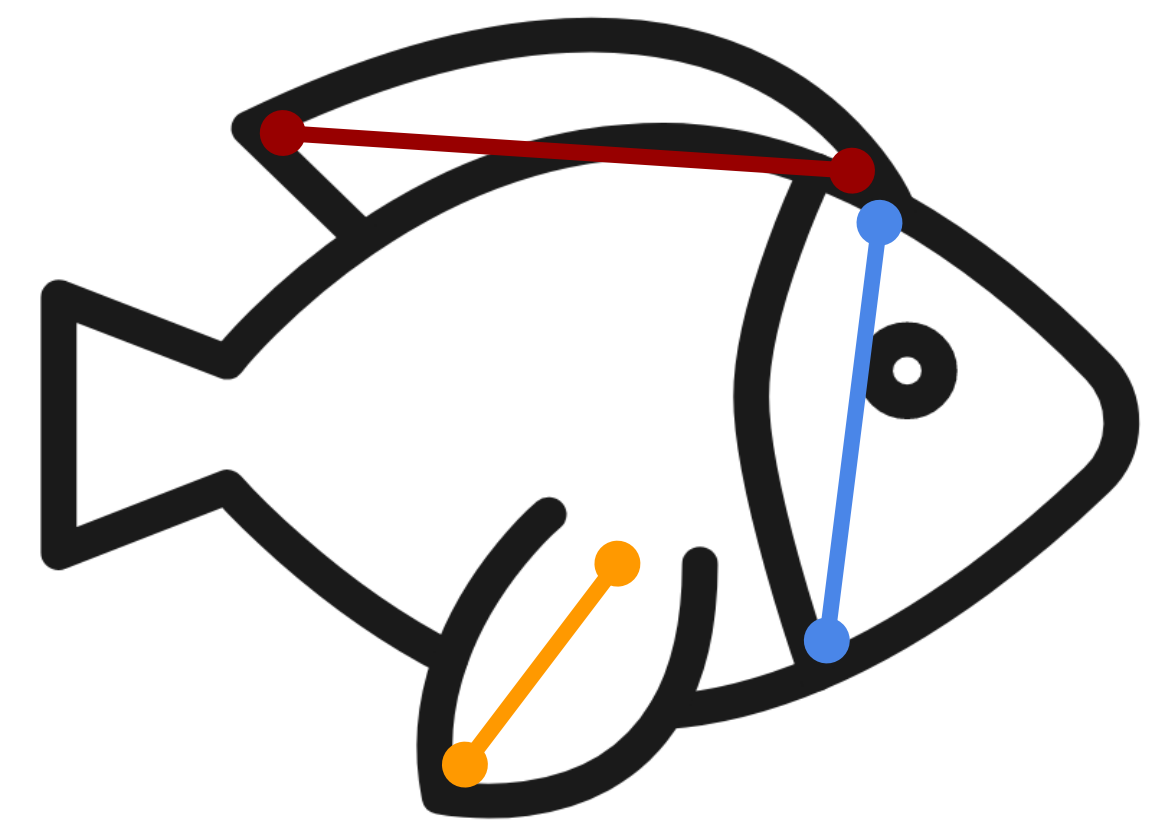
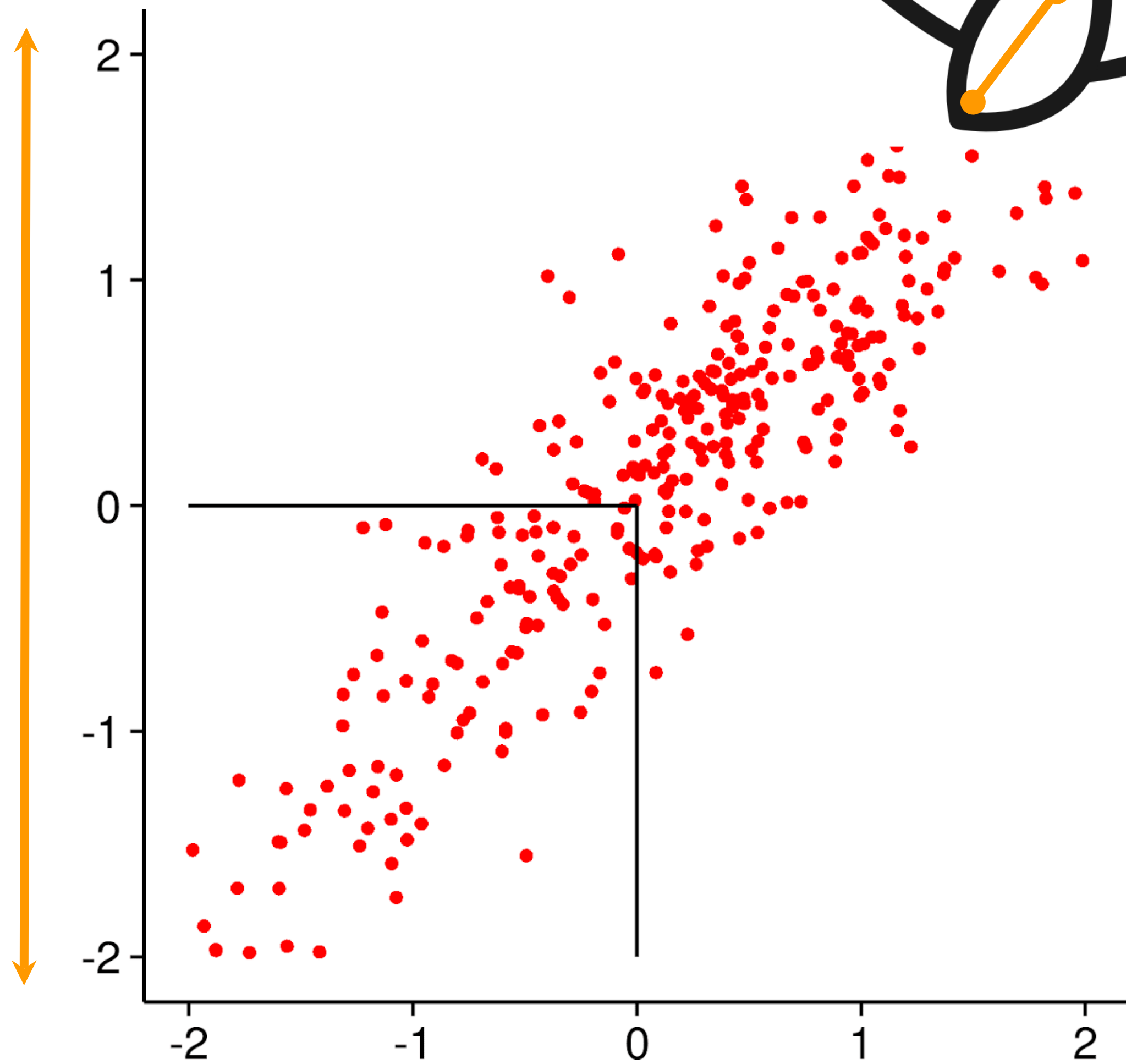
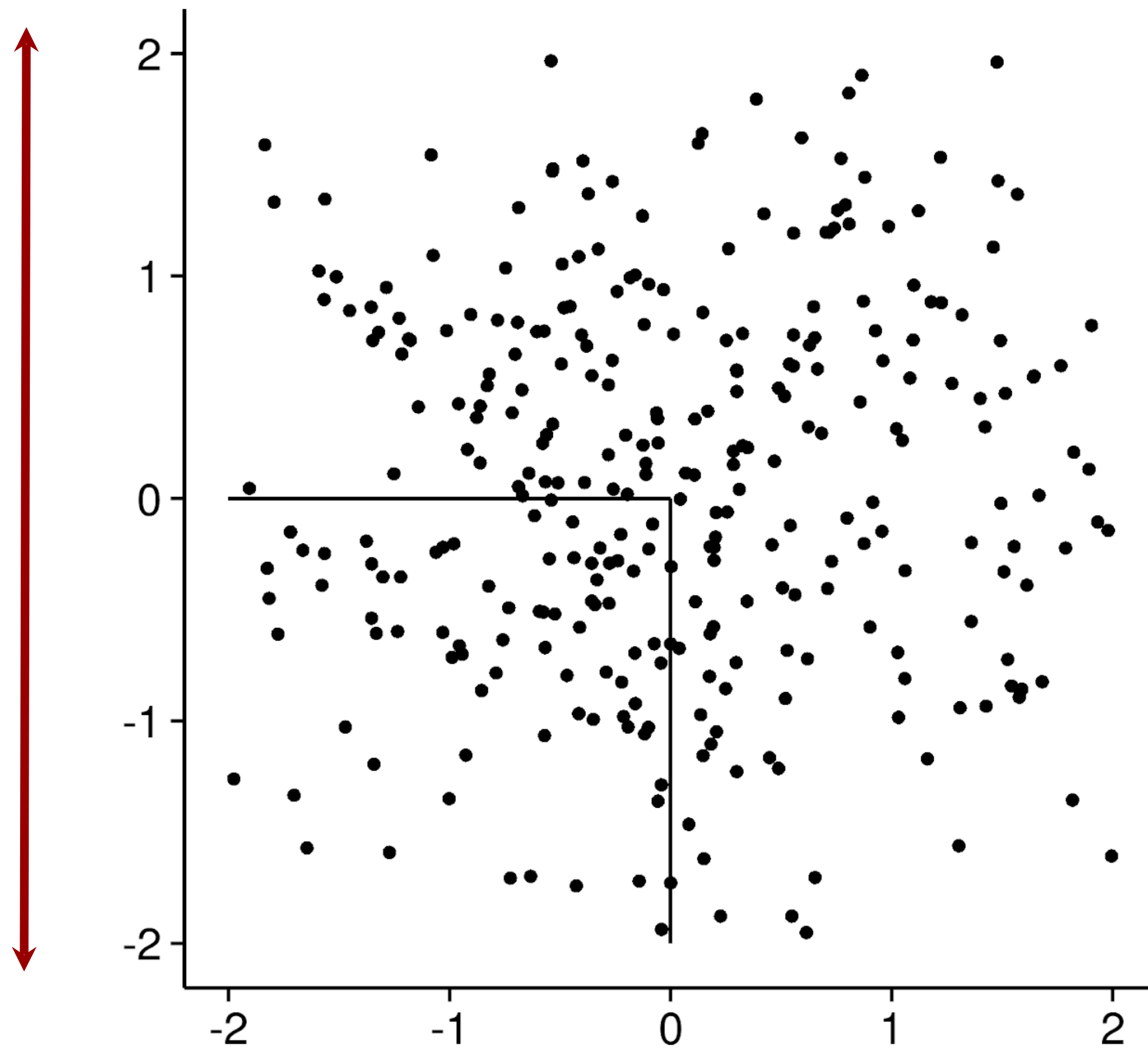




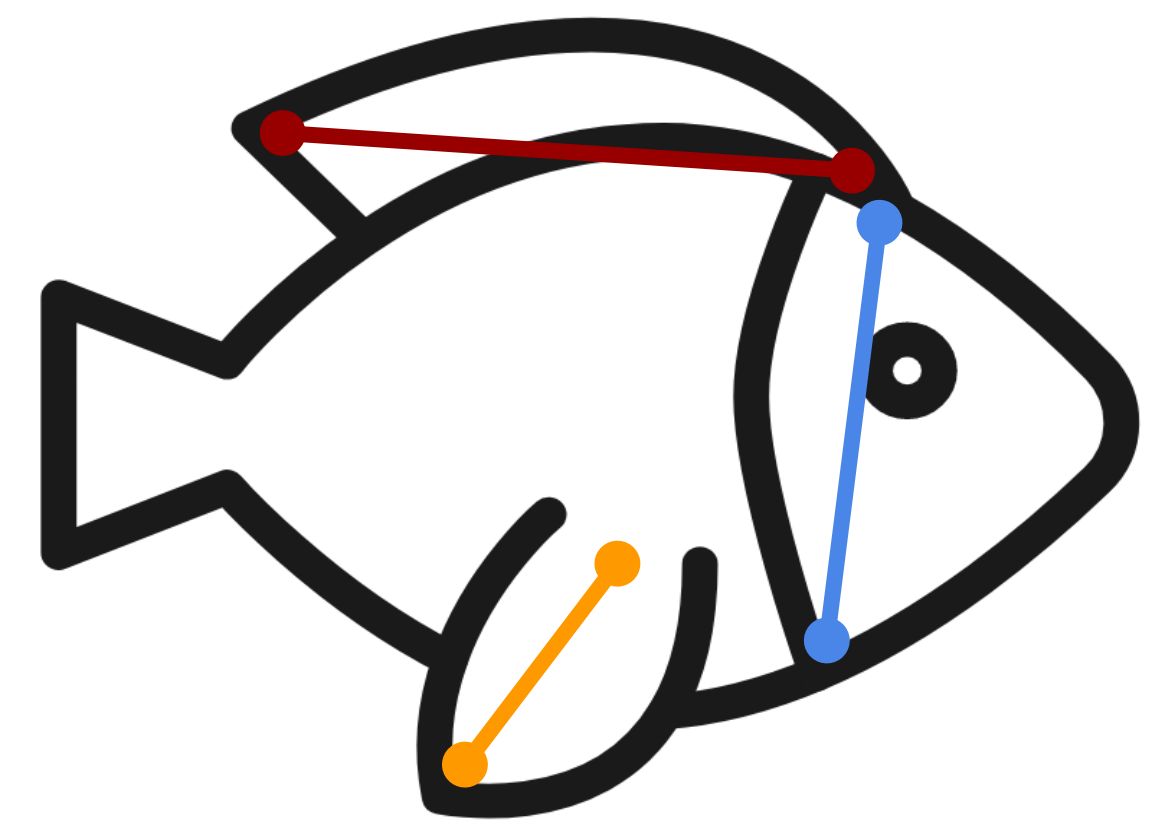
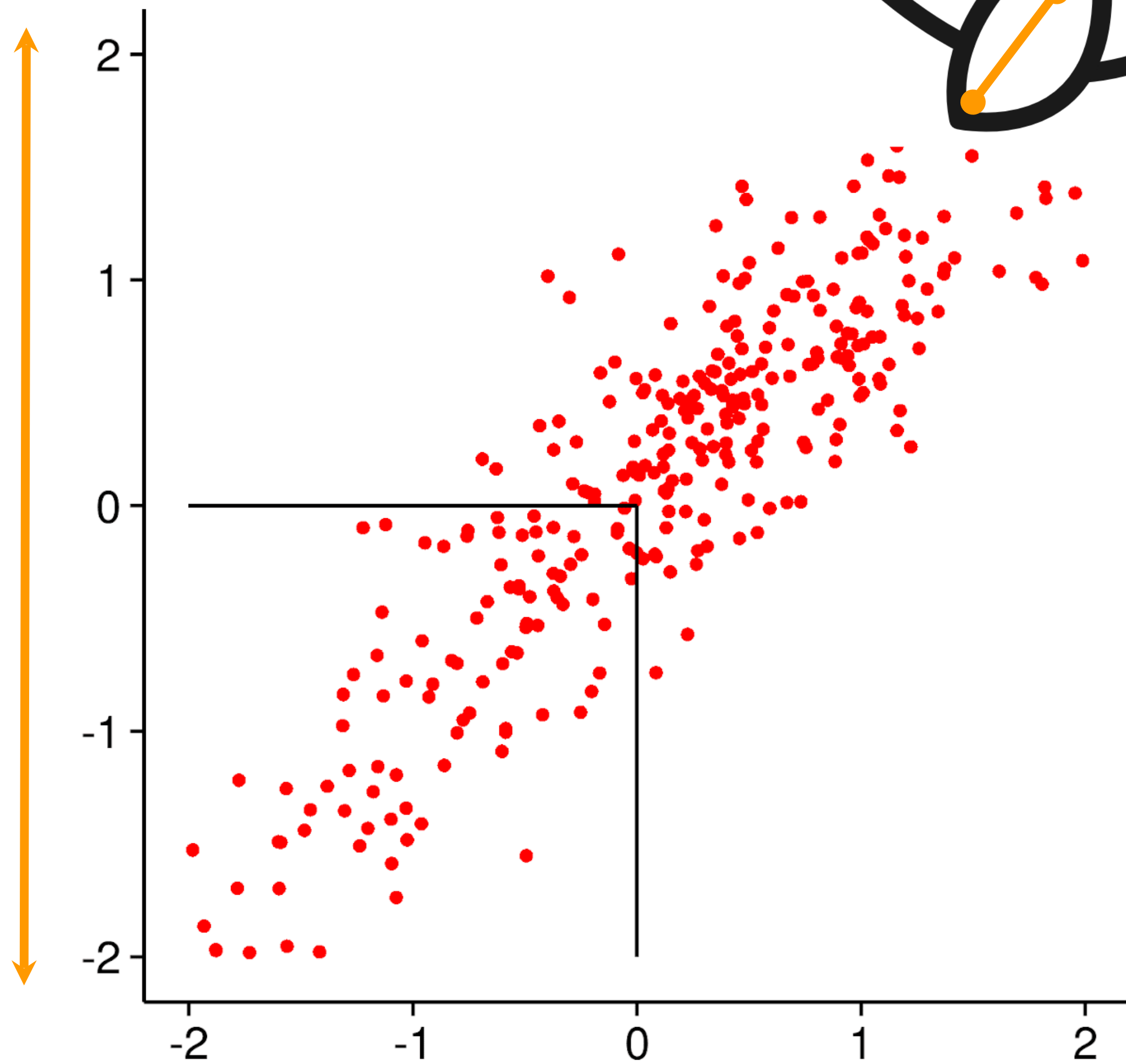
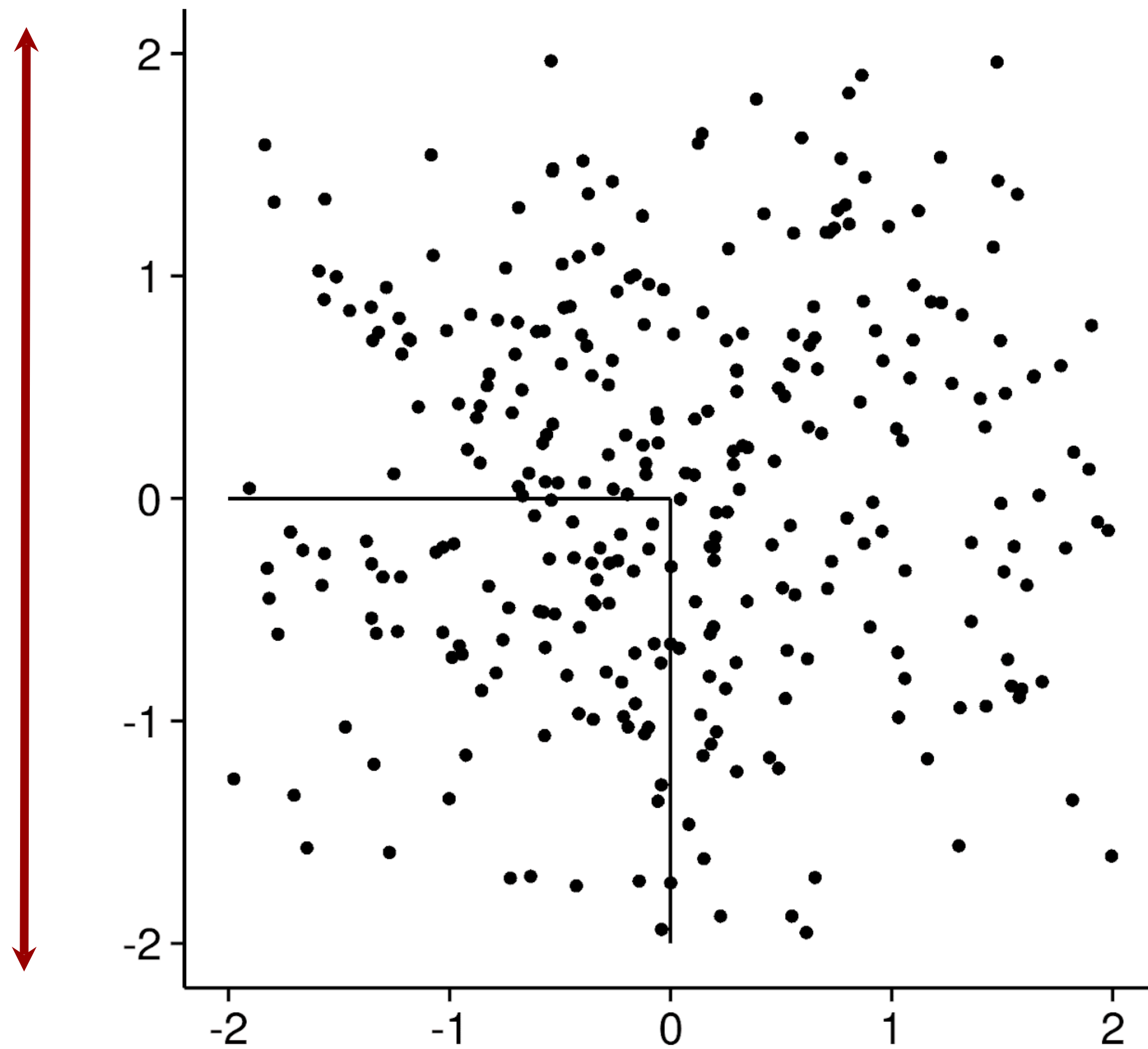


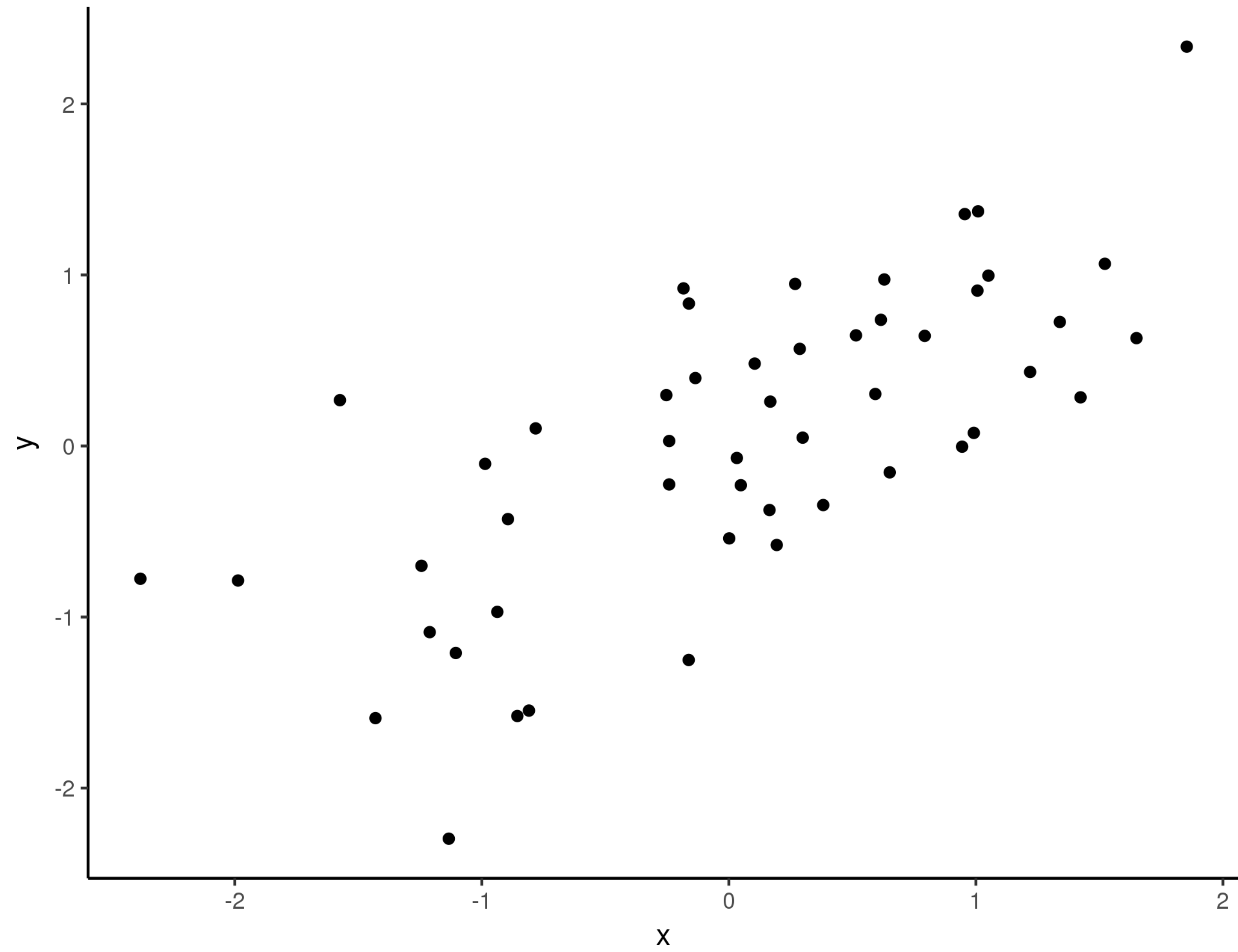


Covariação



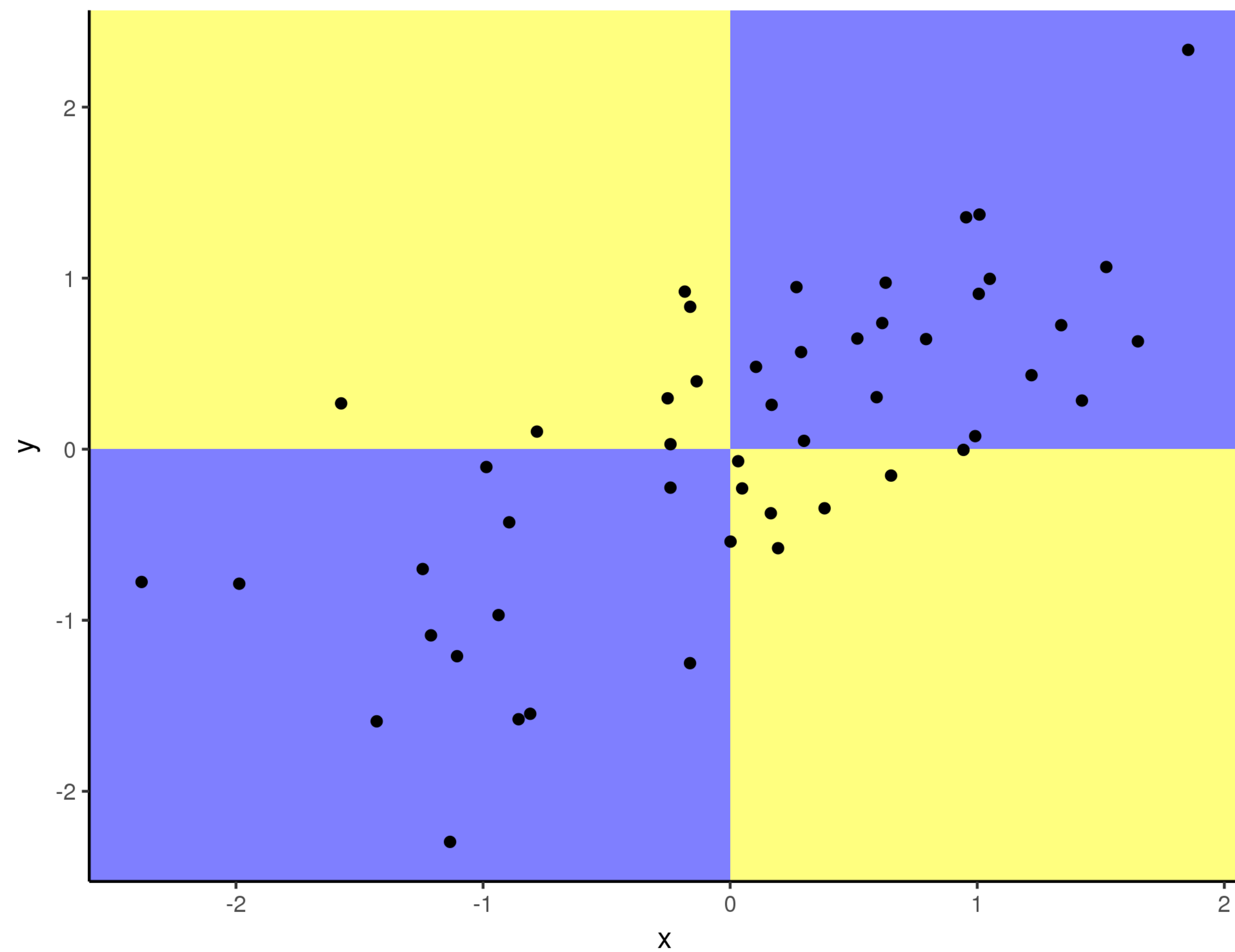
Covariação

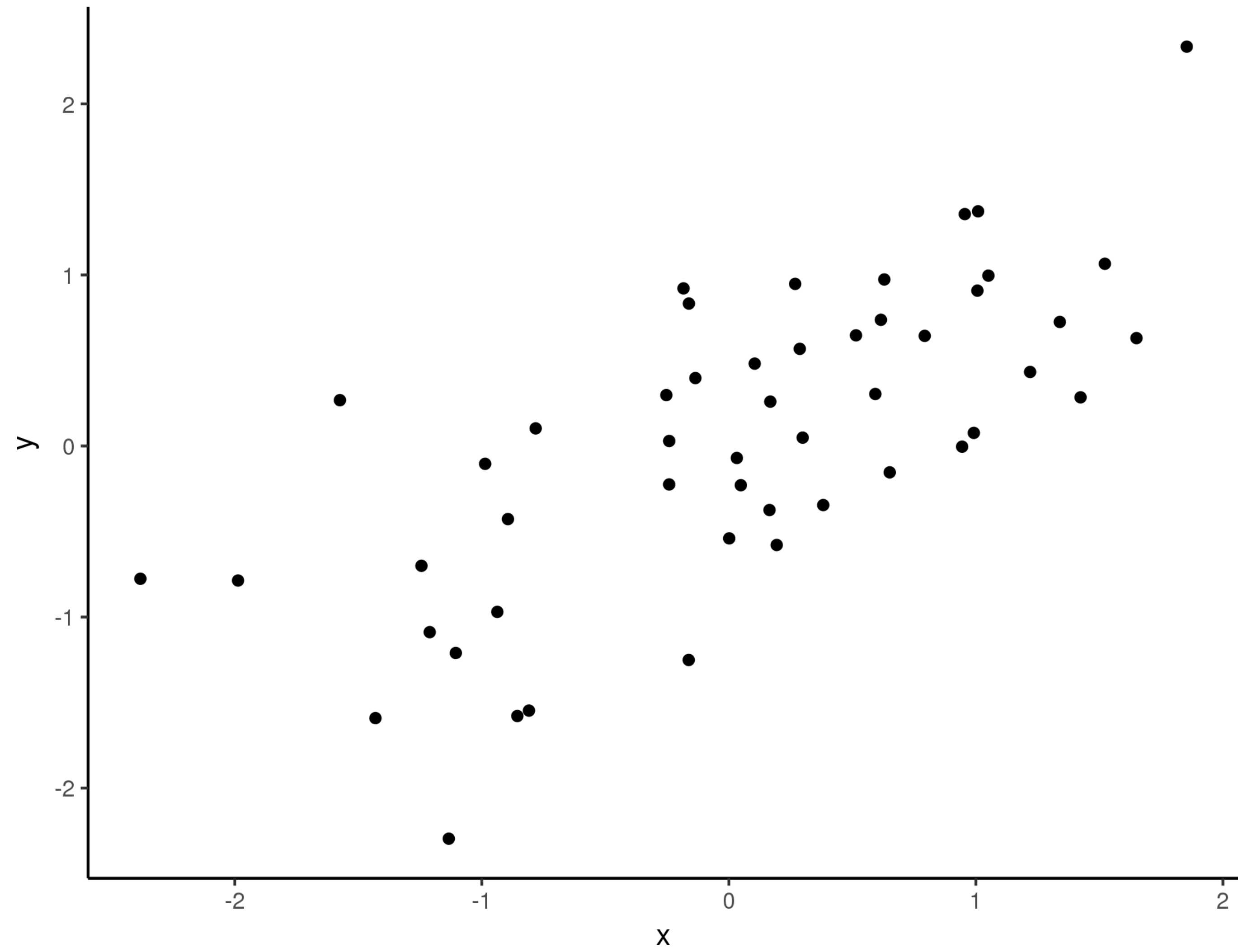


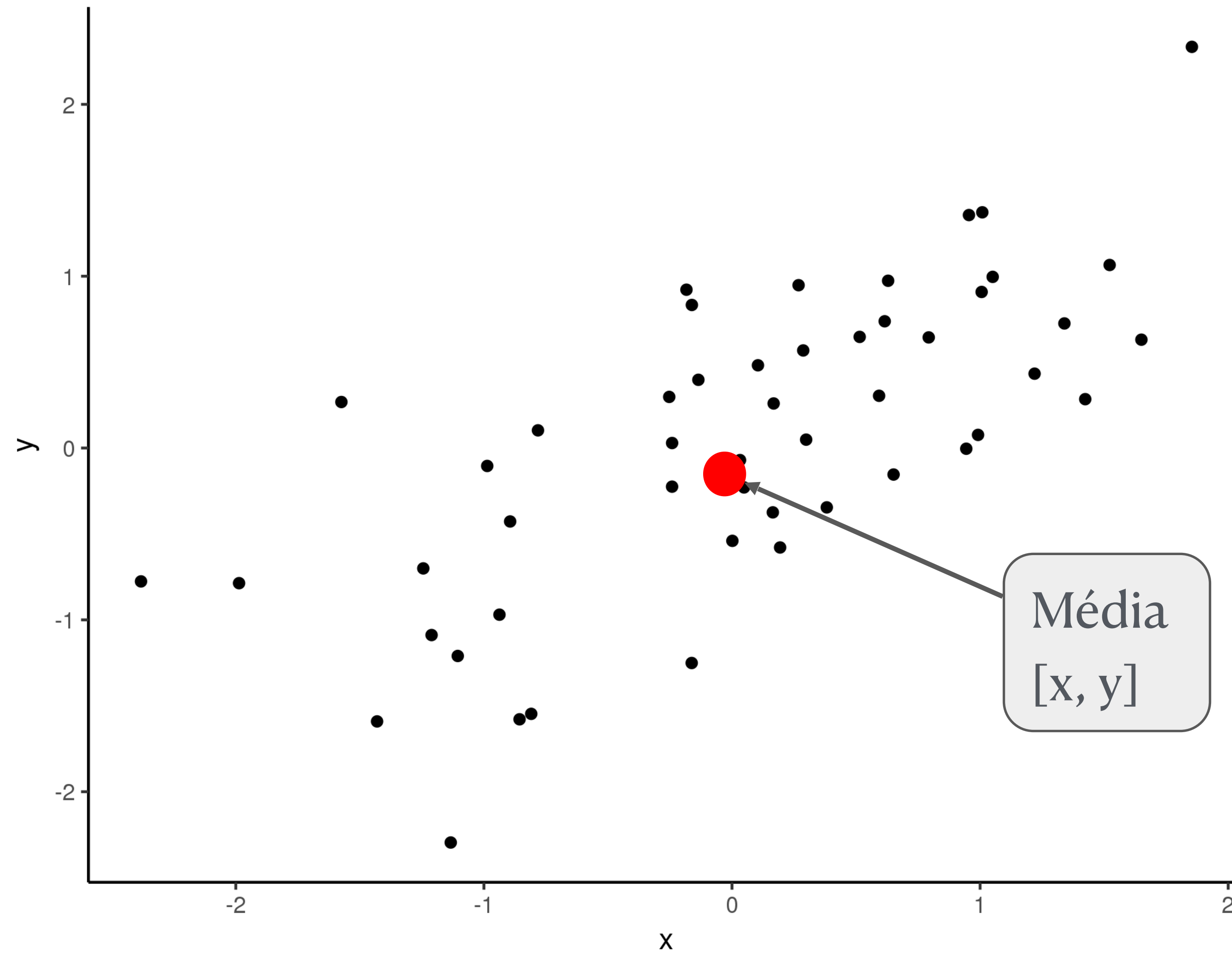


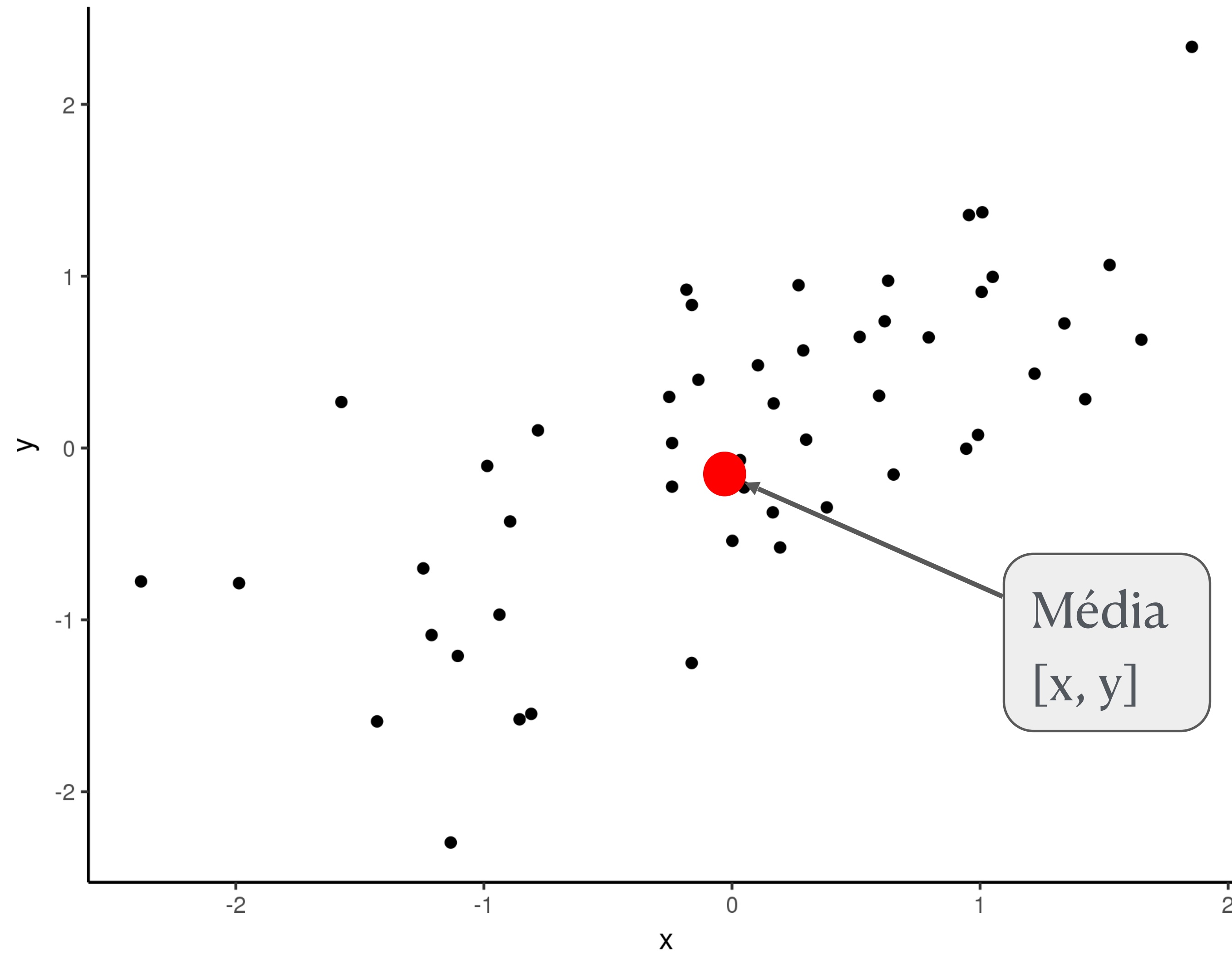
$$\sigma_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})$$

$$\rho_{xy} = \frac{\sigma_{xy}}{\sqrt{\sigma_x^2 \sigma_y^2}}$$

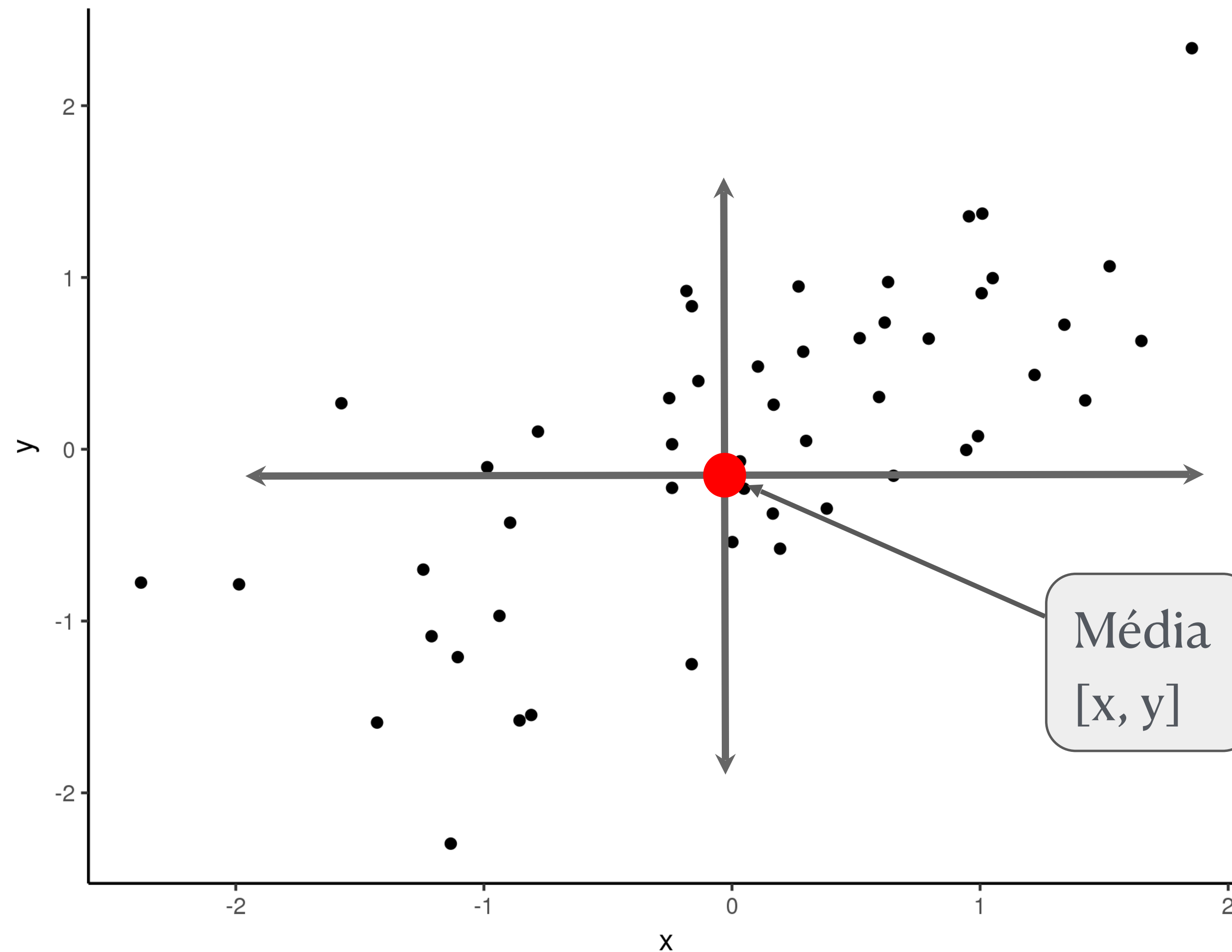








Normal Multivariada

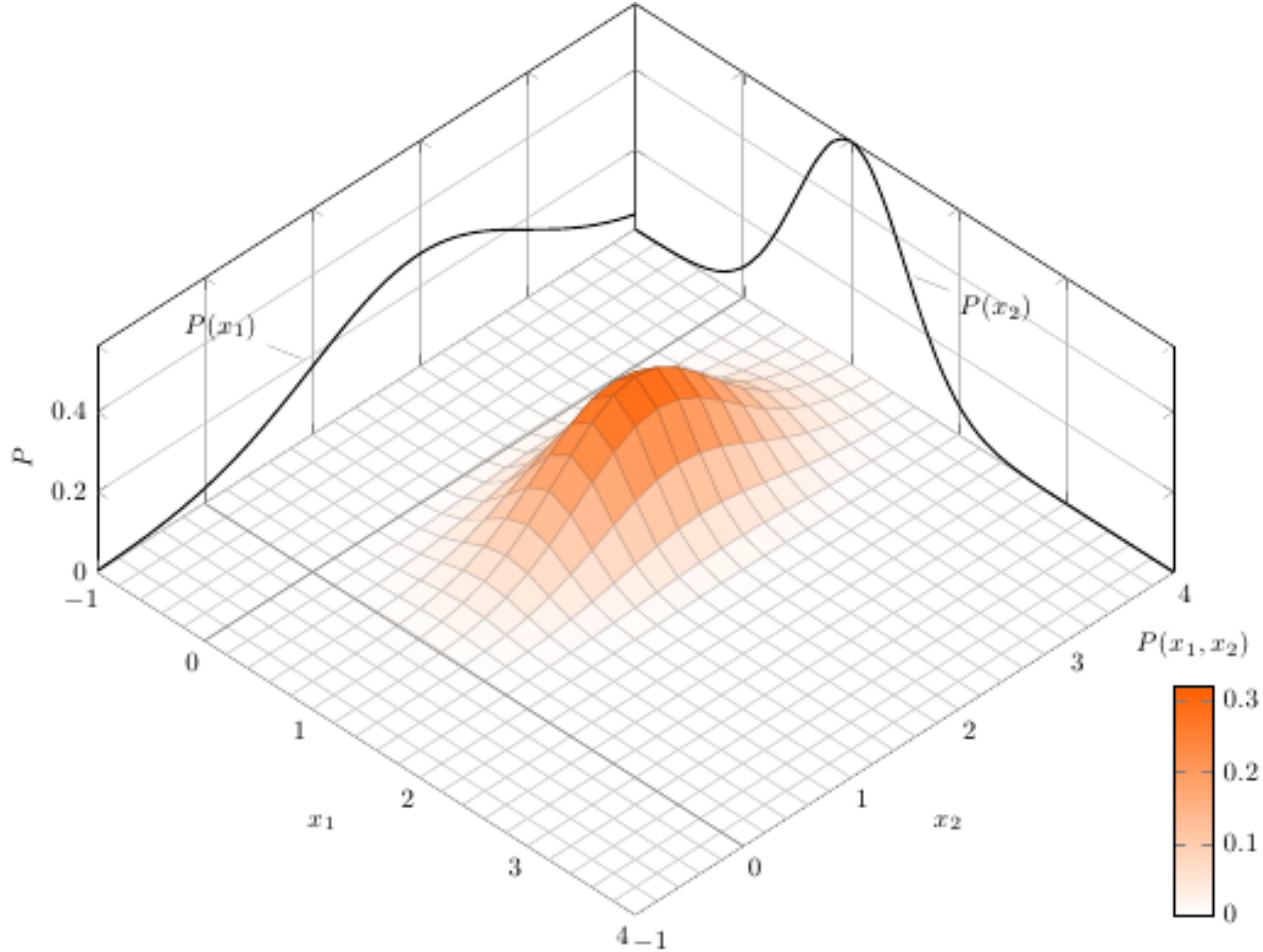
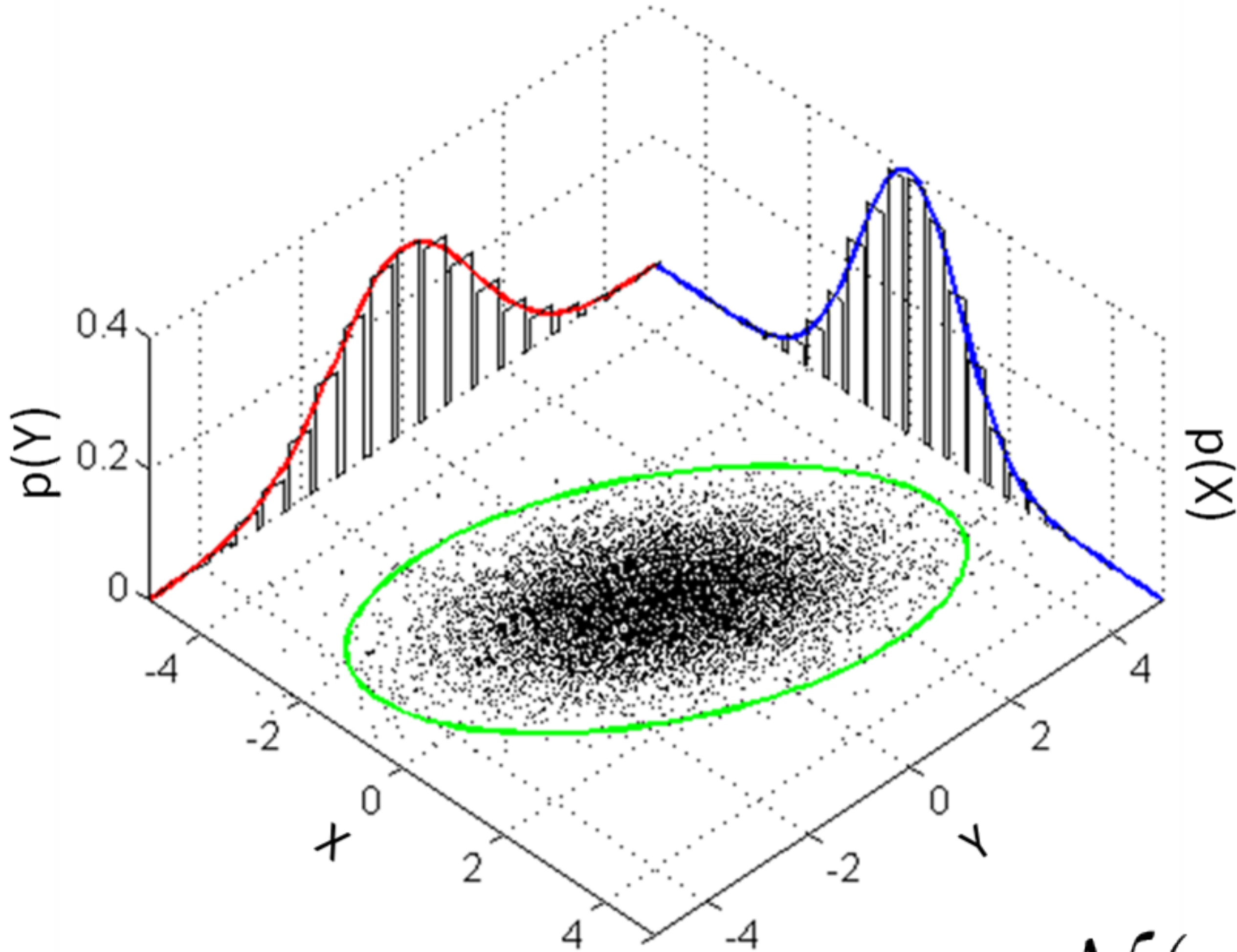


Média
[x, y]

Covariância

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{yx} & \sigma_y^2 \end{bmatrix}$$

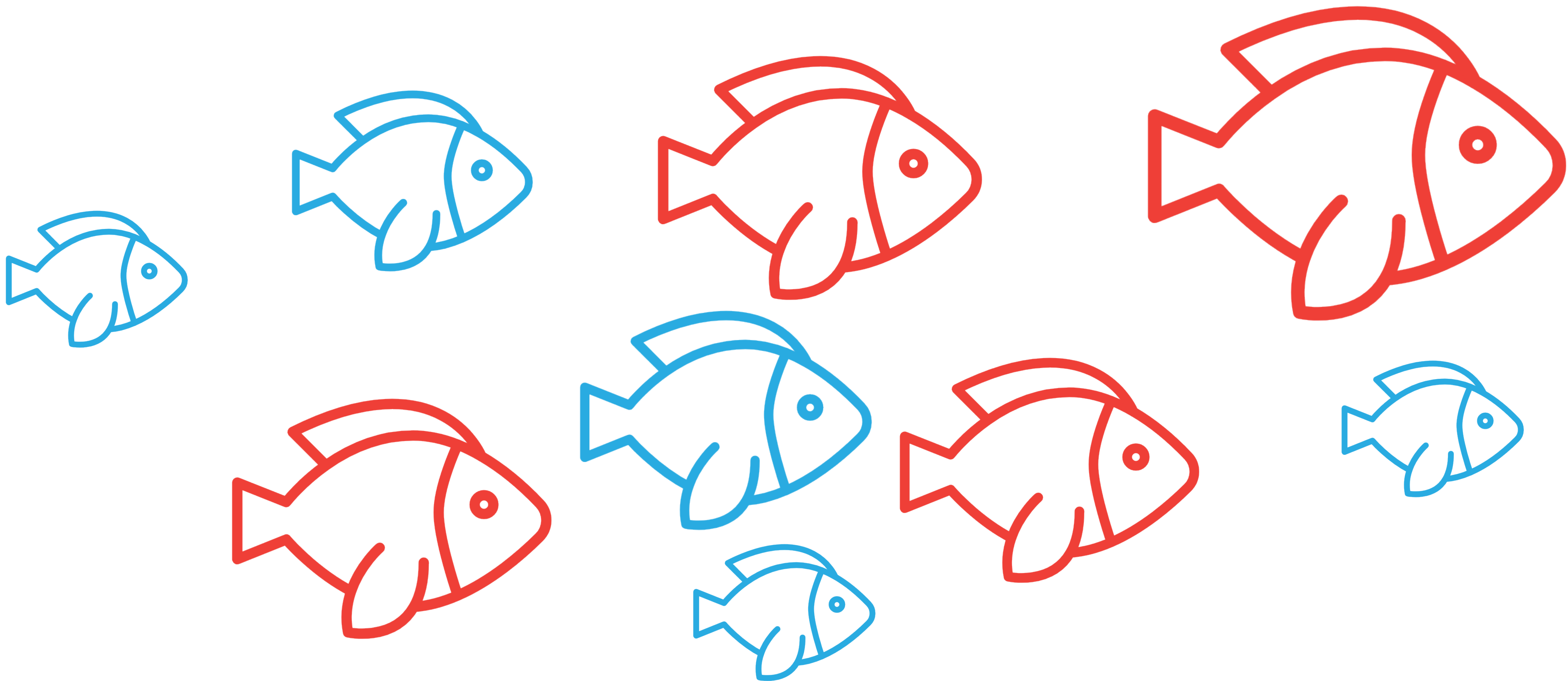
$$\Sigma = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{yx} & \sigma_y^2 \end{bmatrix}$$

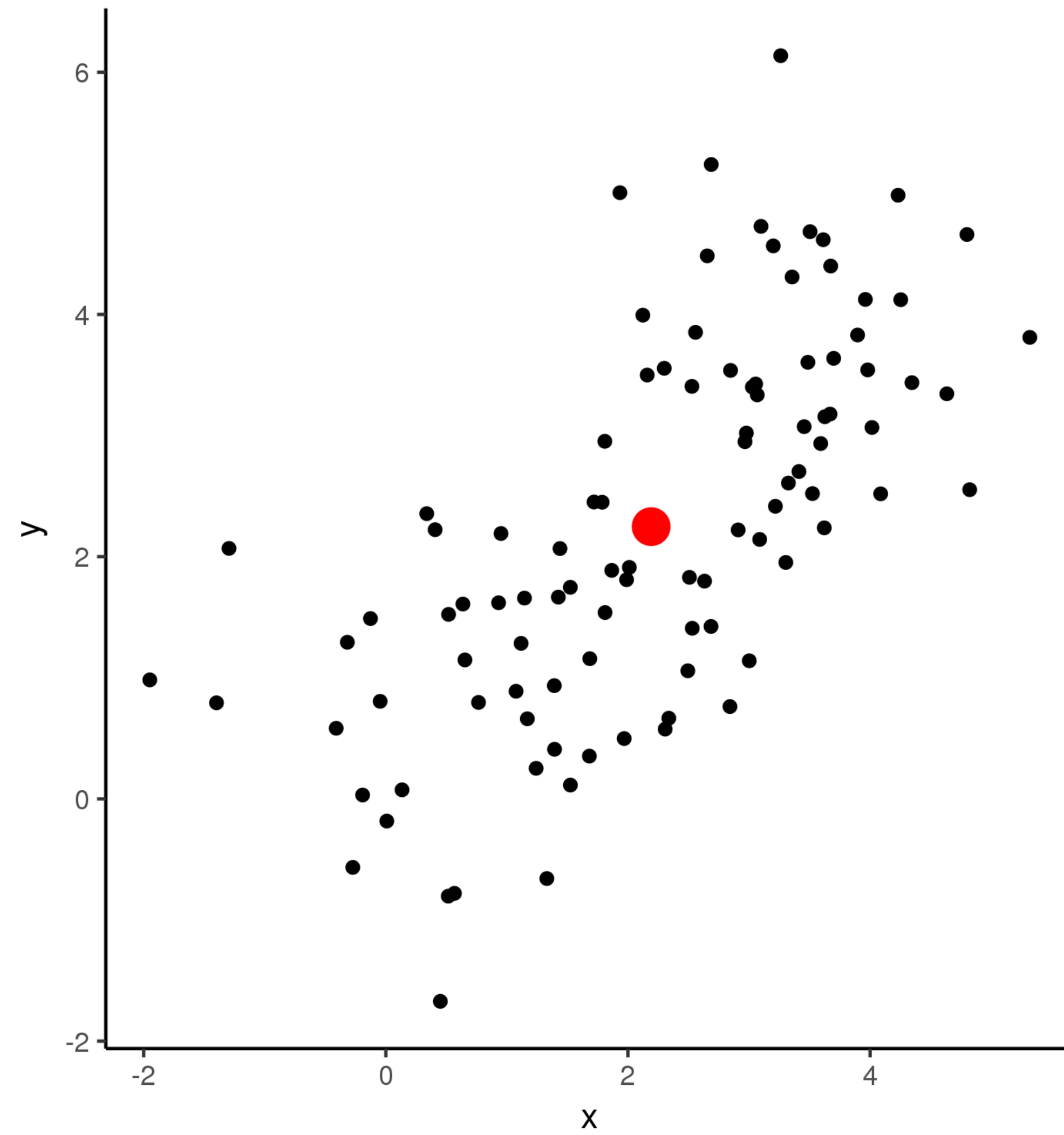


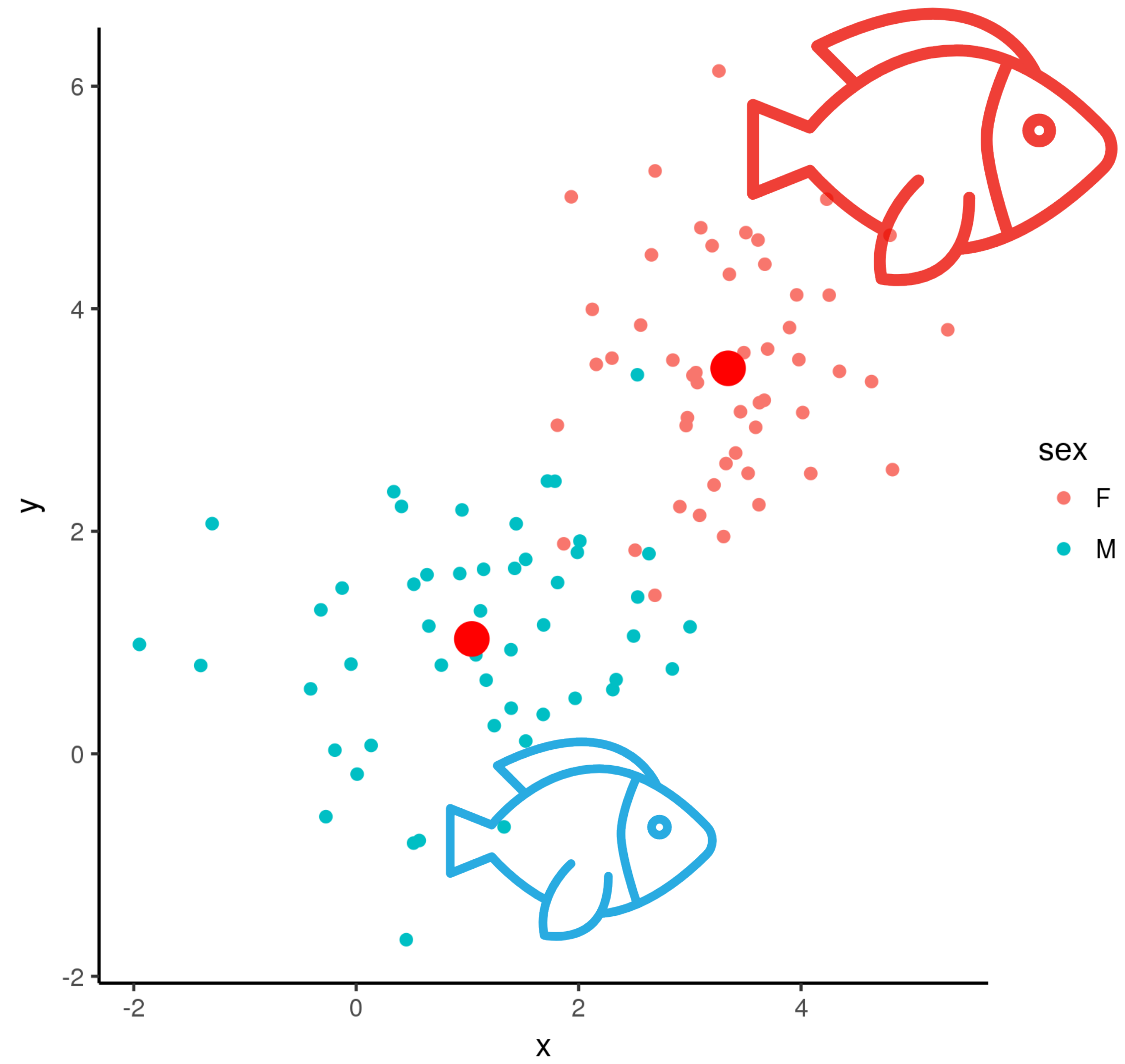
$$\Sigma_{ij} = \sigma_{x^i x^j}$$

$$\mathbf{x}_i = \{x_i^1, x_i^2, \dots, x_i^p\}$$

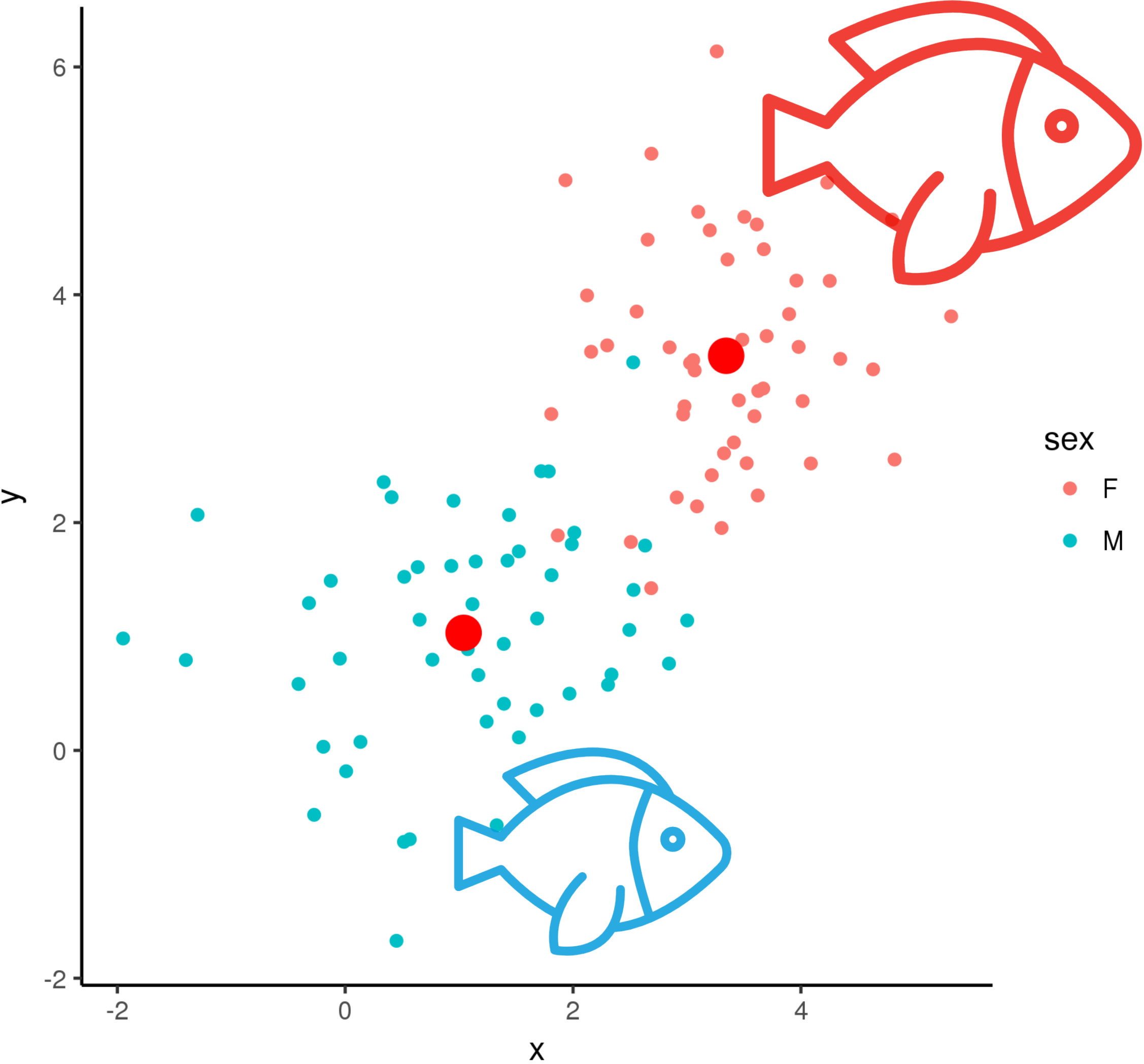
$$\mathcal{N}(\mathbf{x}_i|\mu, \Sigma) \approx e^{\frac{1}{2}(\mathbf{x}_i-\mu)^T \Sigma^{-1}(\mathbf{x}_i-\mu)}$$



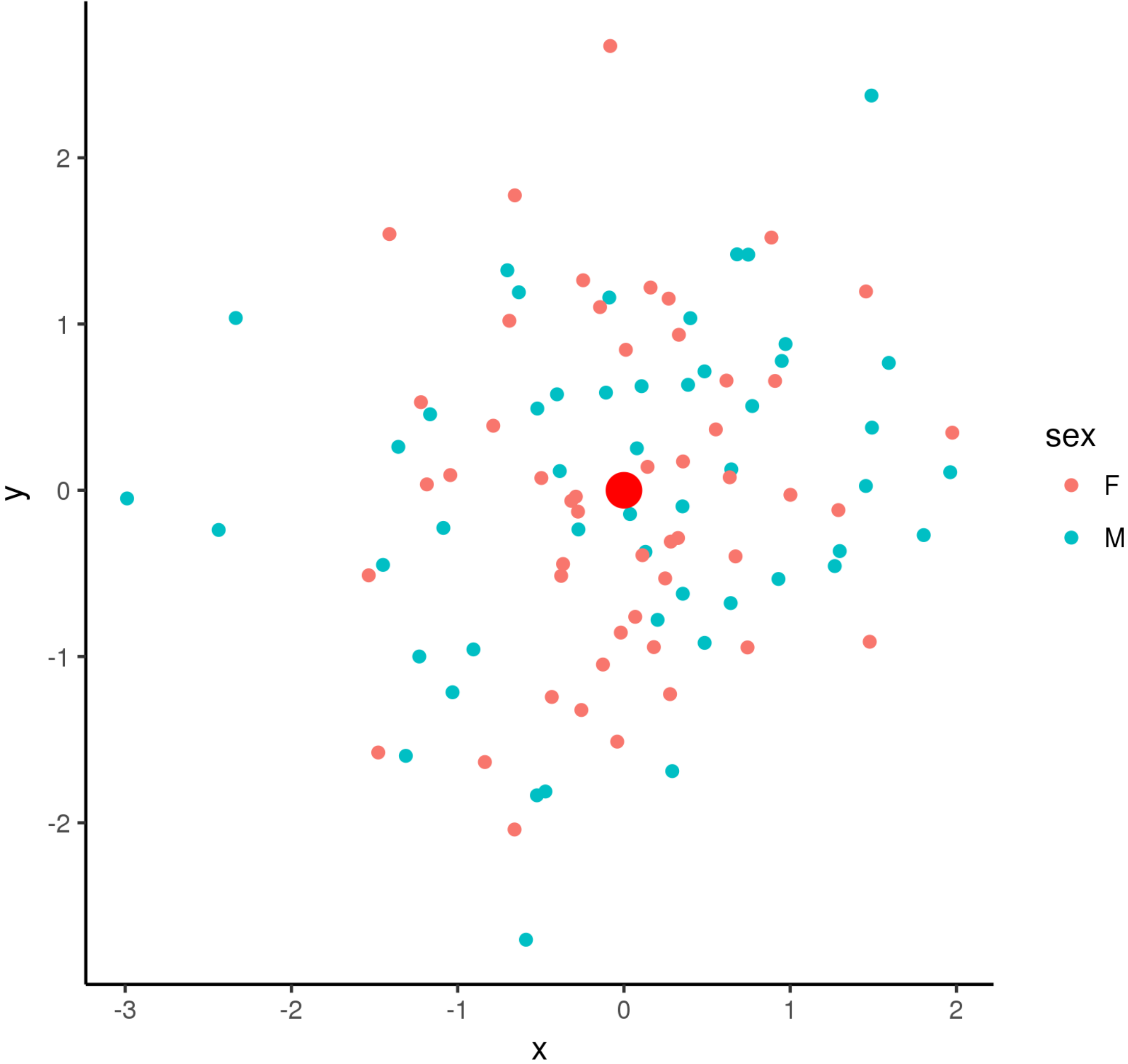




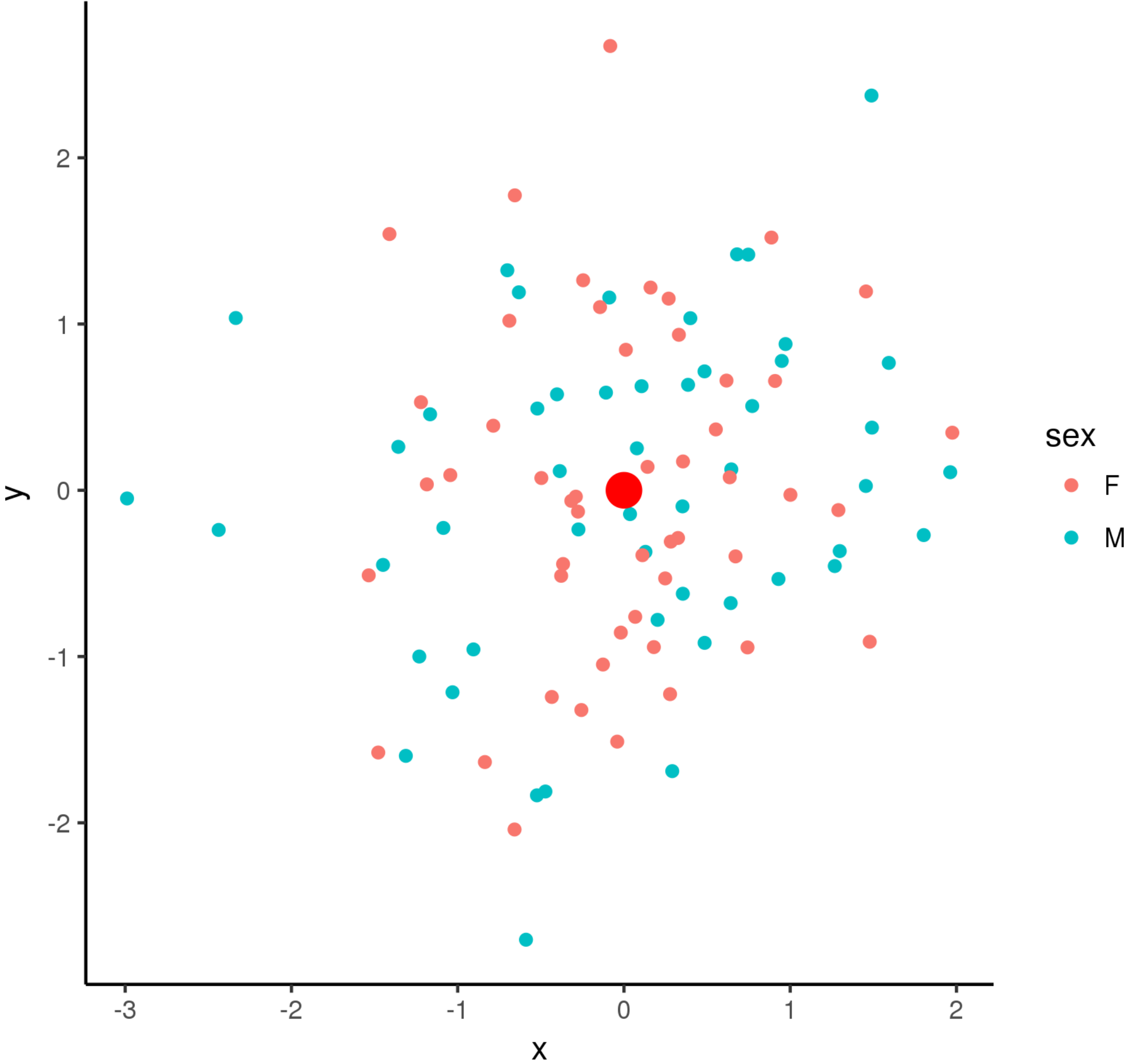
$$\sigma_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x}_{g[i]}) (y_i - \bar{y}_{g[i]})$$



$$\sigma_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x}_{g[i]}) (y_i - \bar{y}_{g[i]})$$

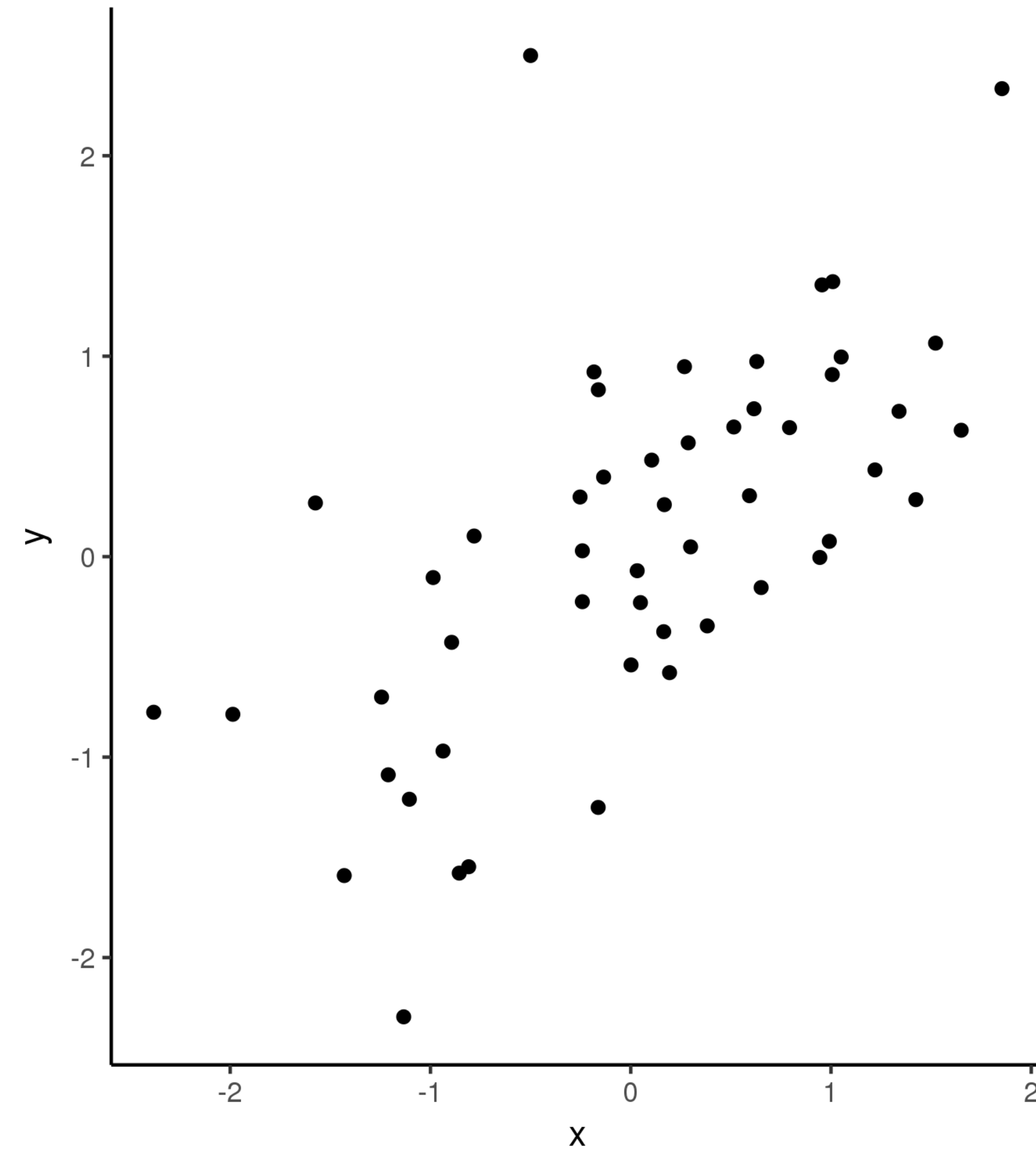


$$\sigma_{xy} = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x}_{g[i]}) (y_i - \bar{y}_{g[i]})$$

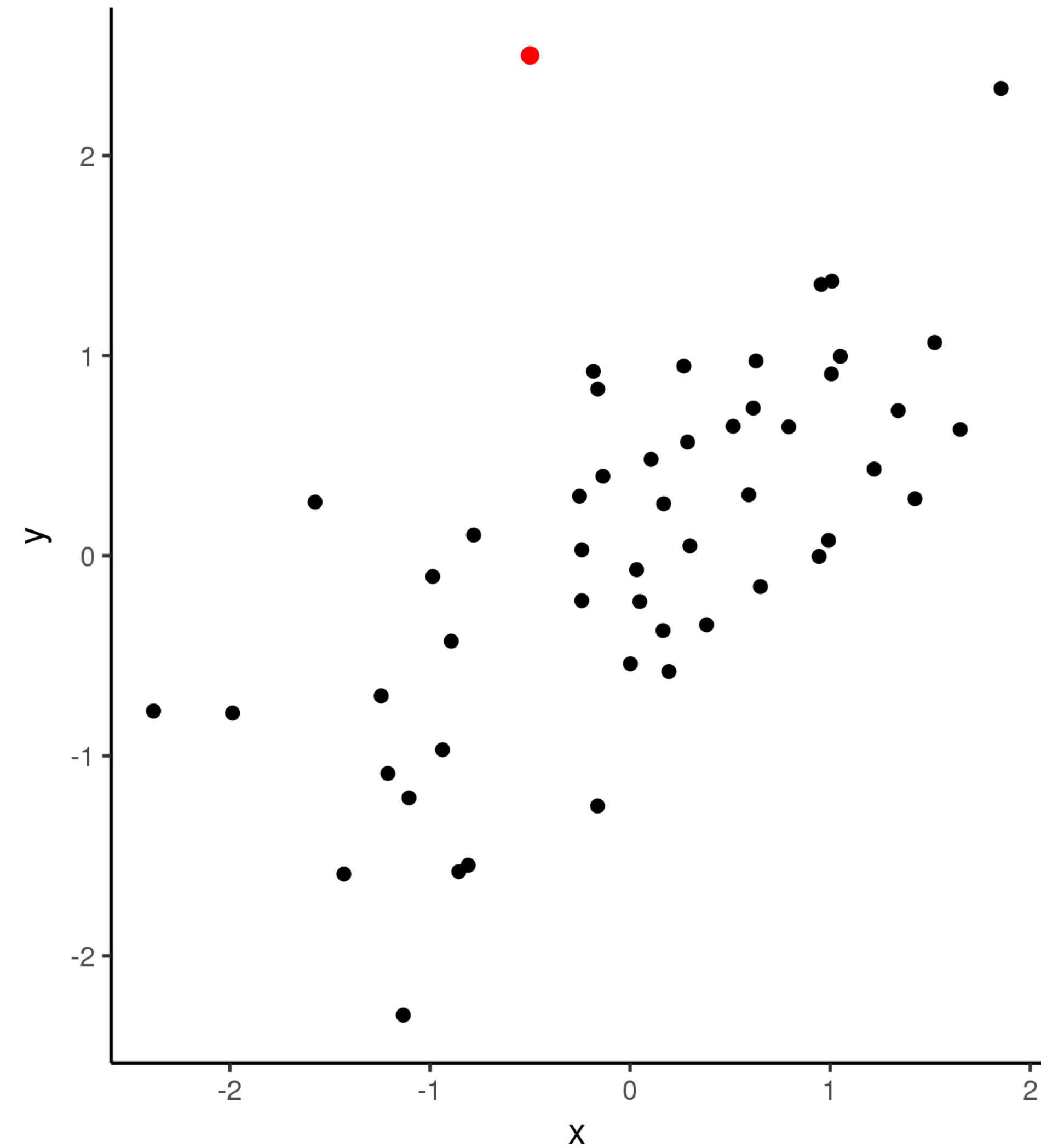


Complicações multivariadas

Outliers multivariadas

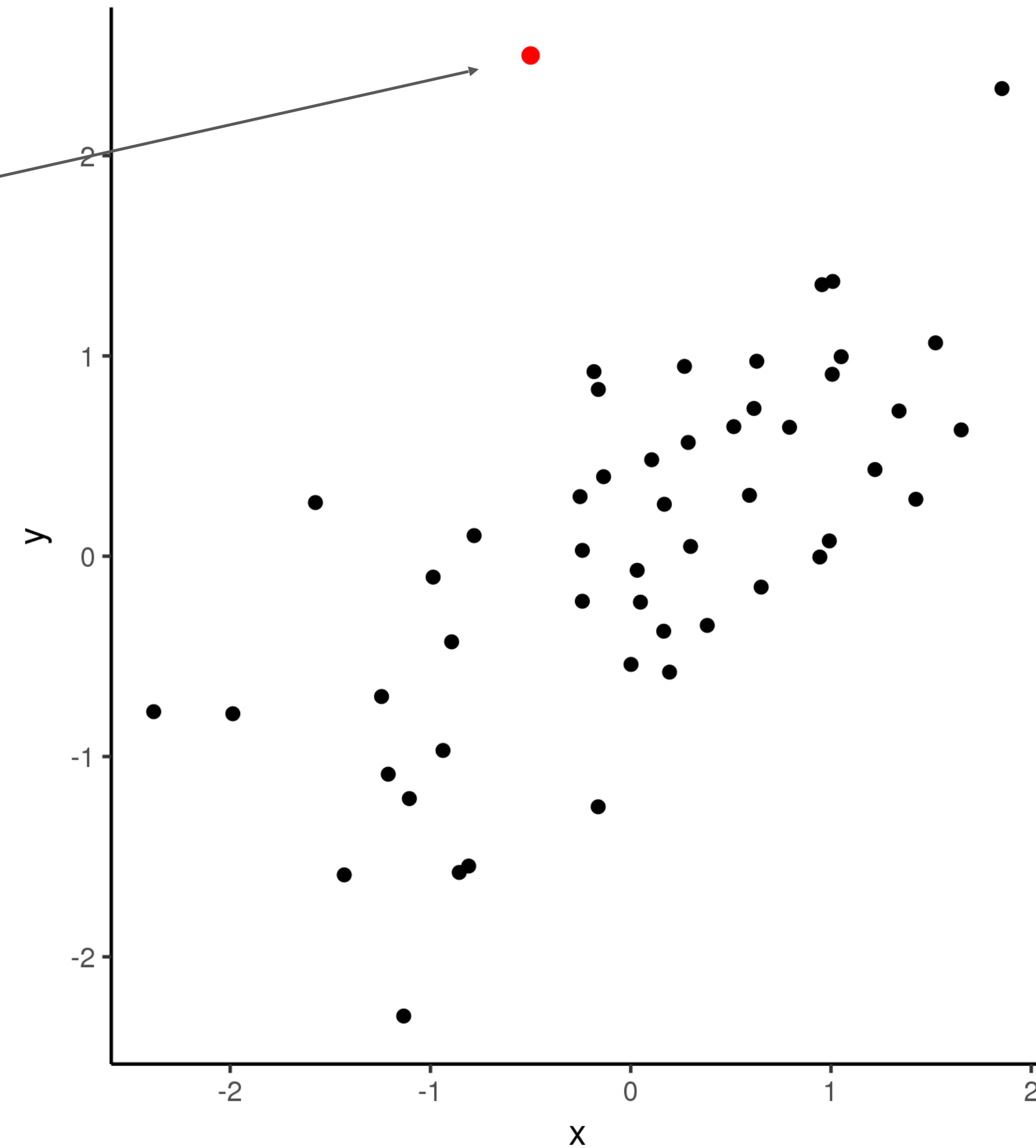


Outliers multivariadas



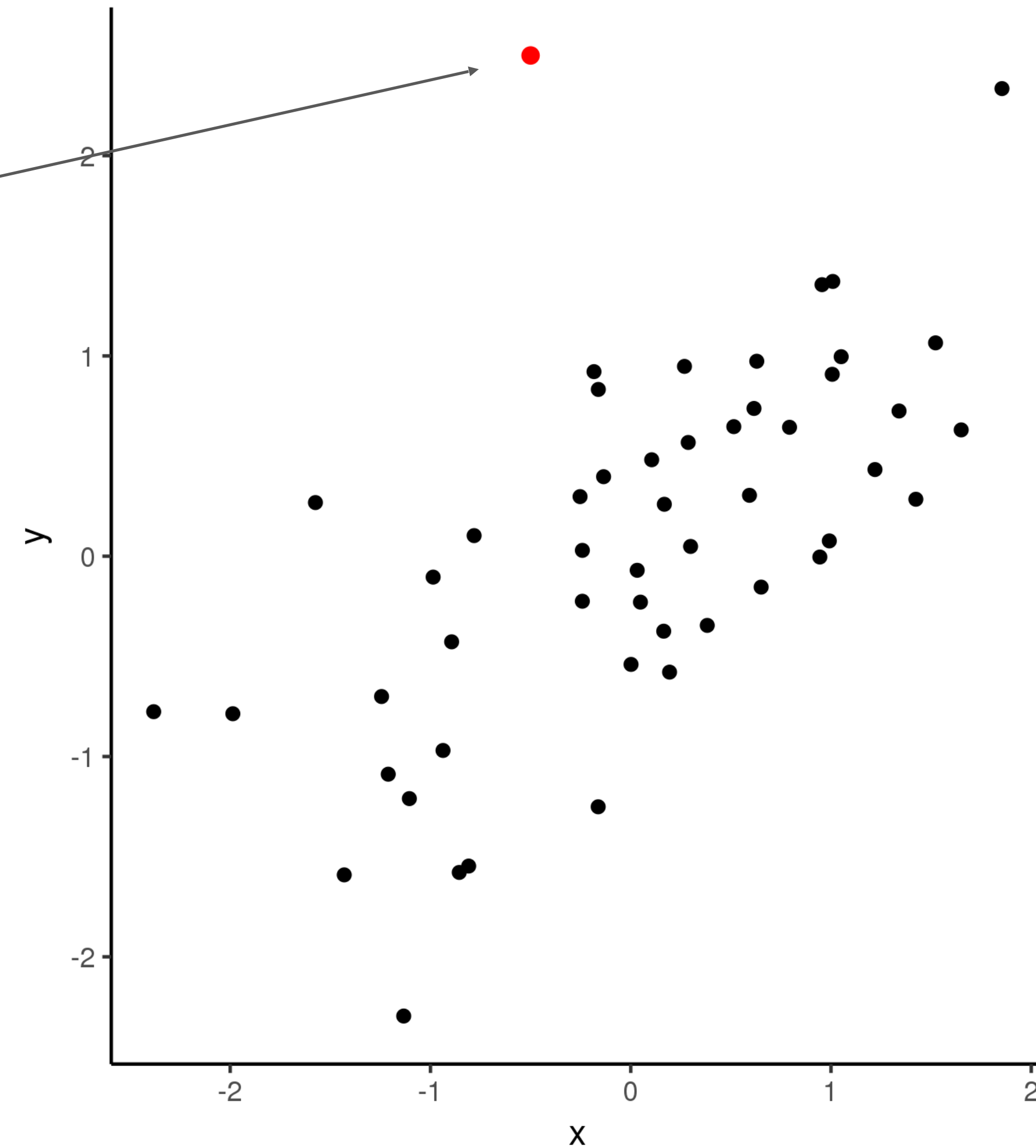
Outliers multivariadas

Por quê
esse?



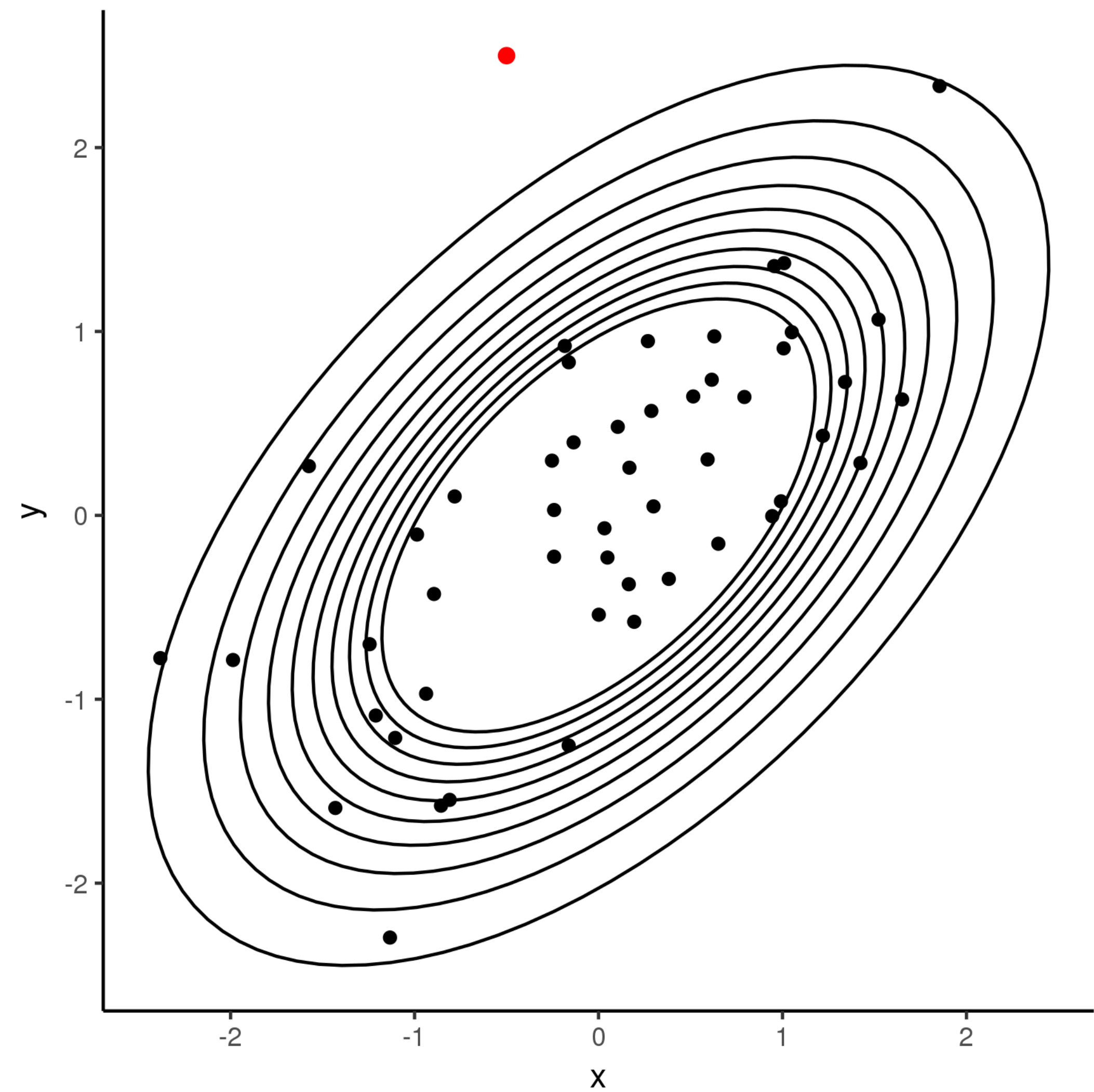
Outliers multivariadas

Por quê
esse?



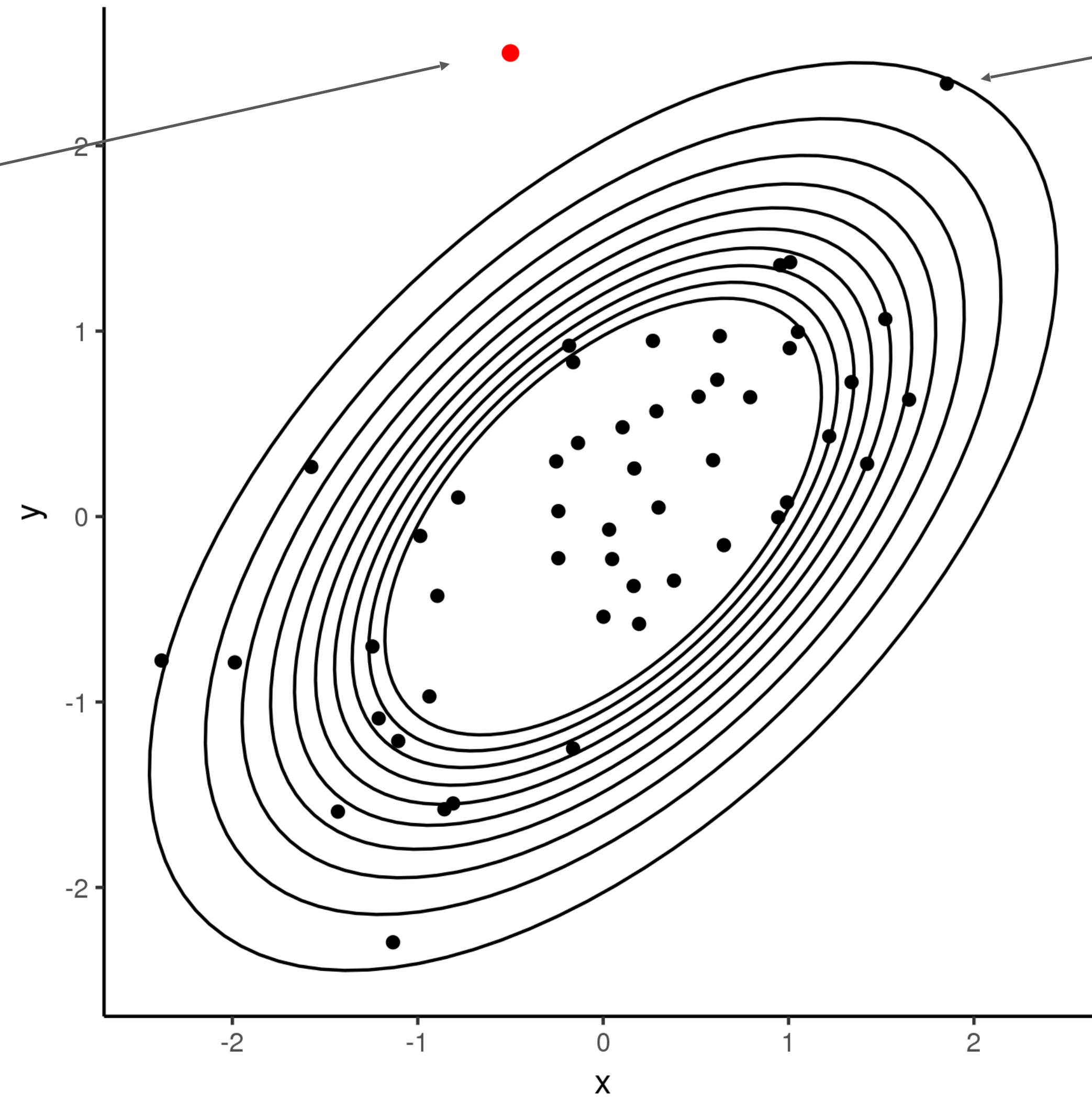
E não esse?

Quantis em 2D



Quantis em 2D

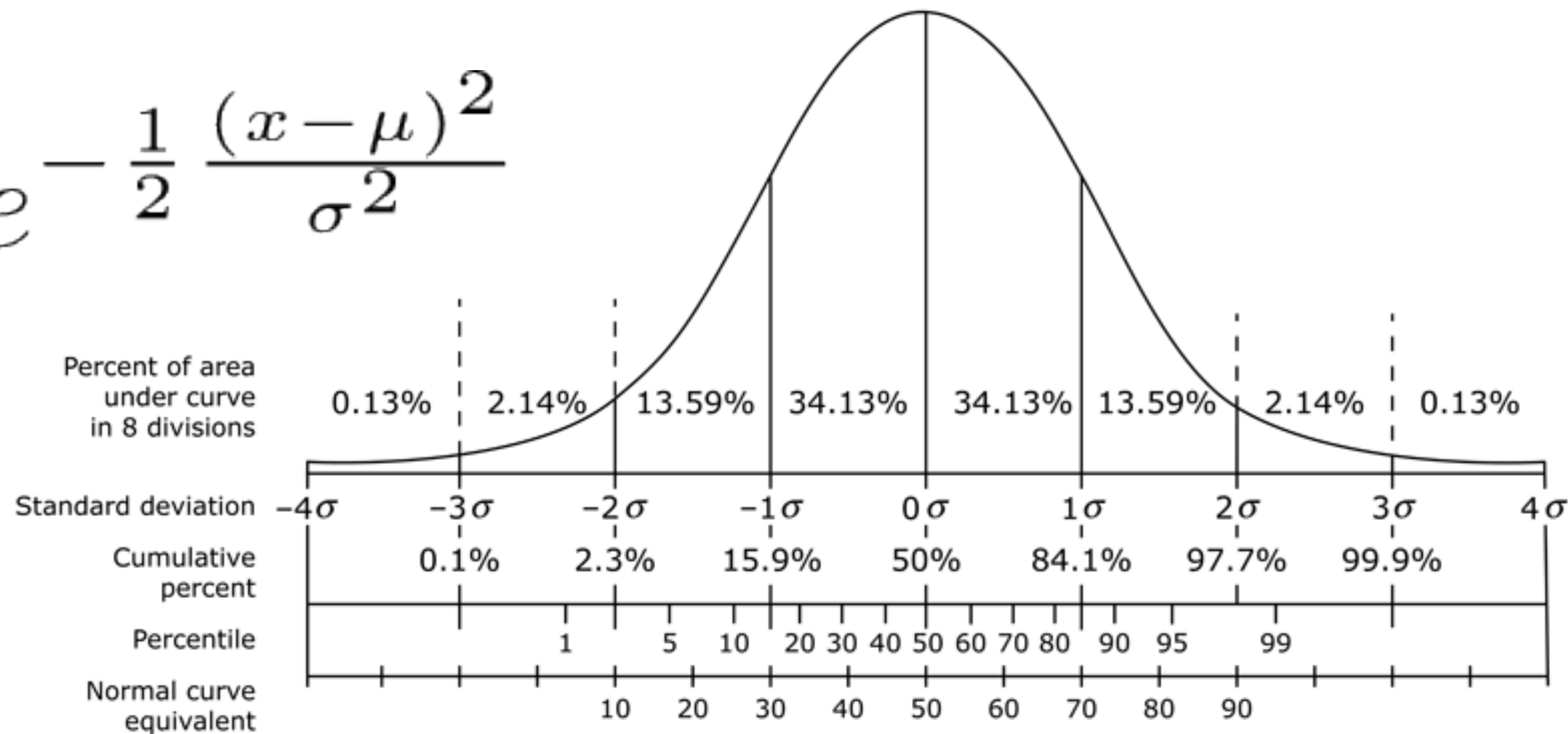
Bem mais longe...



Dentro do
quantil 95%

Unidades de desvio padrão

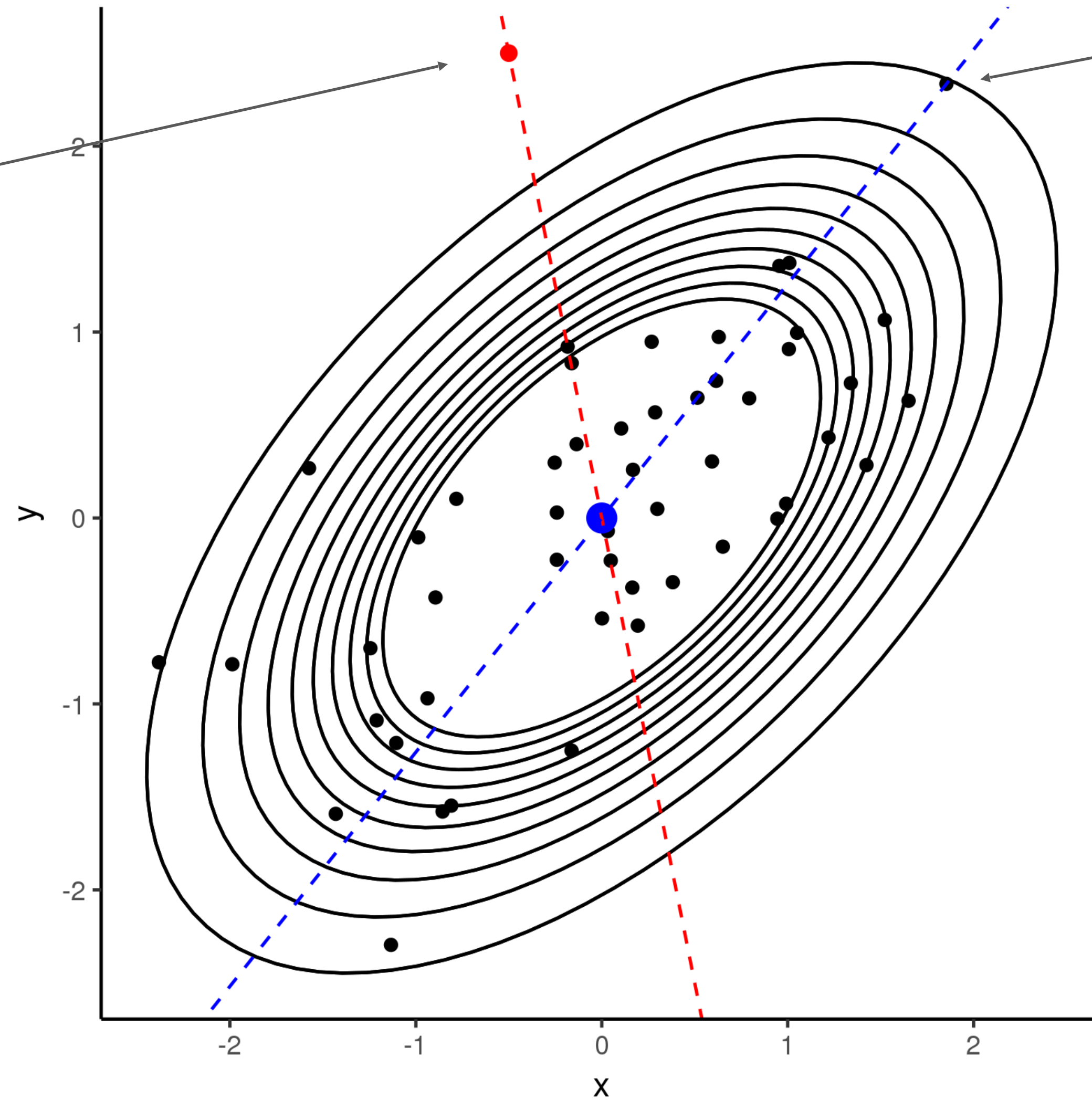
$$e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$



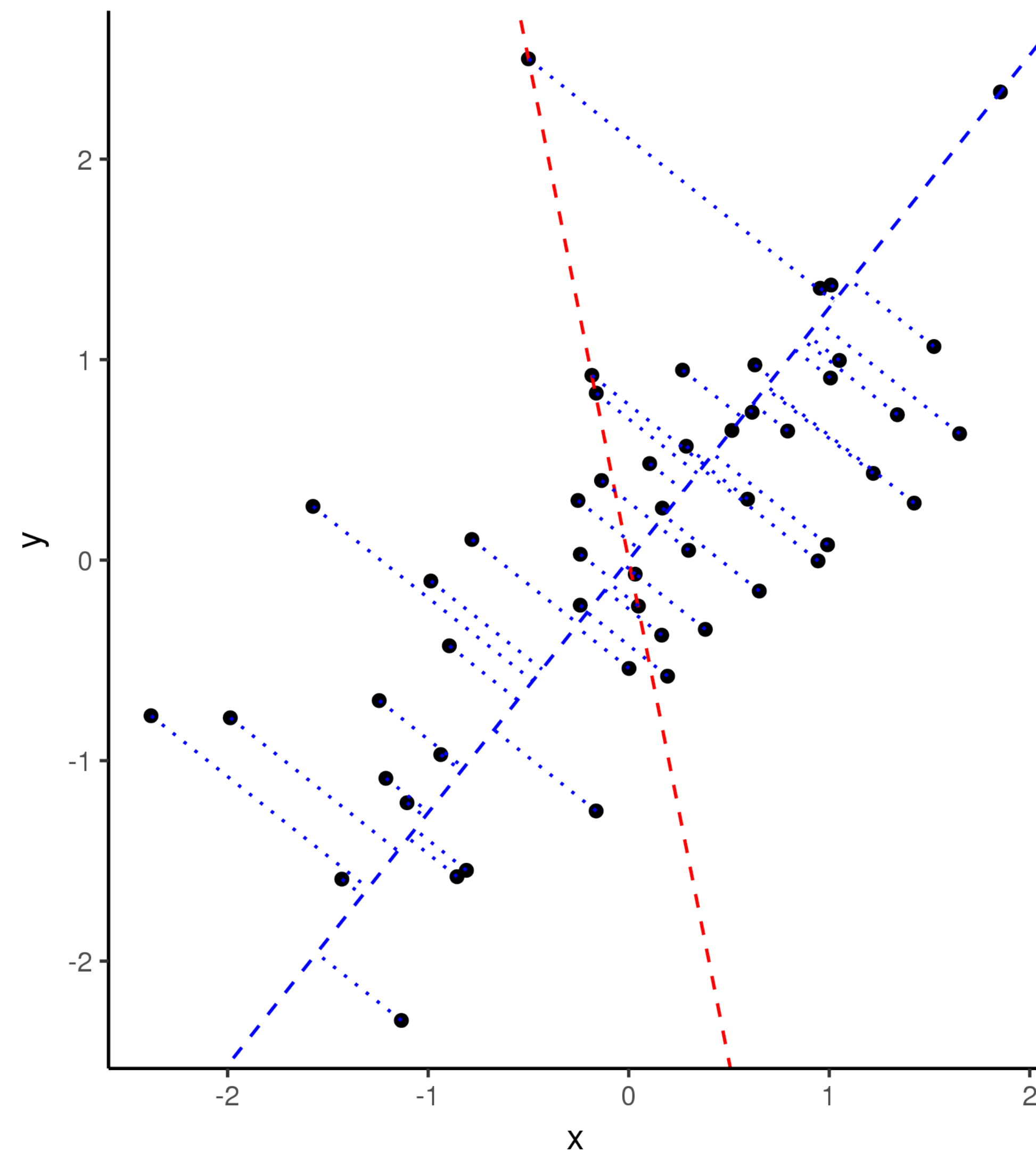
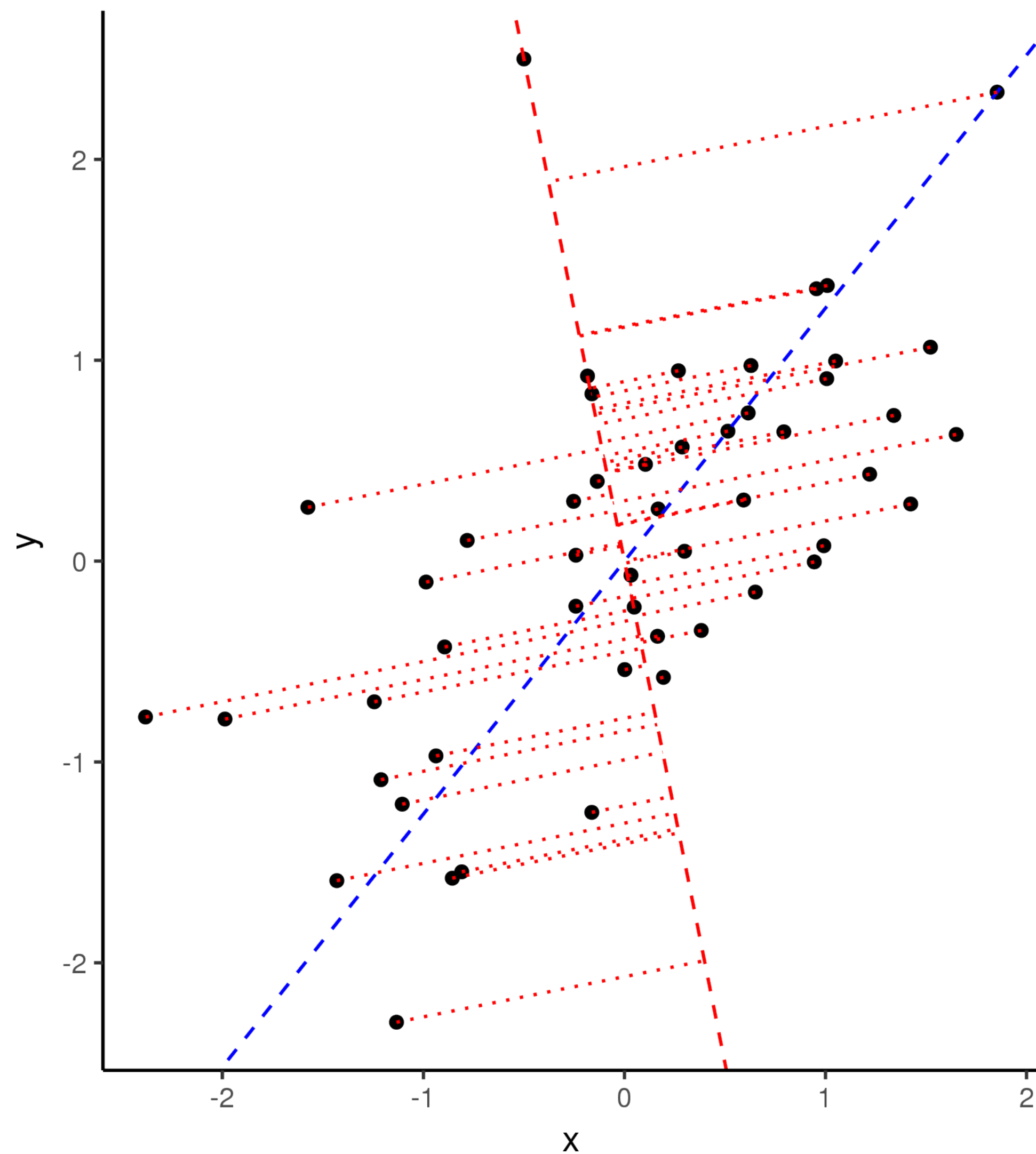
Direções em várias dimensões

Bem mais longe...

Dentro do
quantil 95%

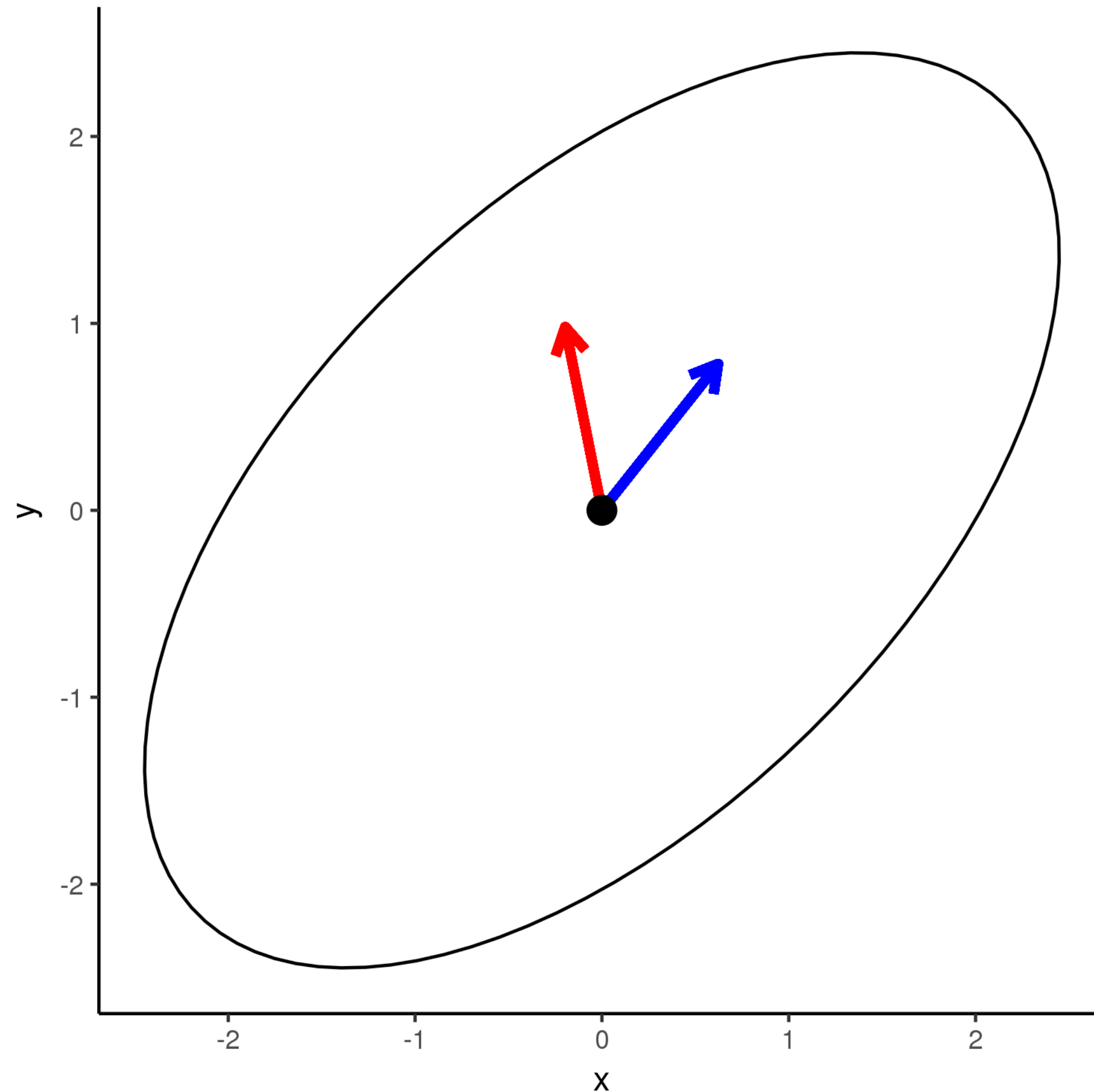


Projeções e variação direcional



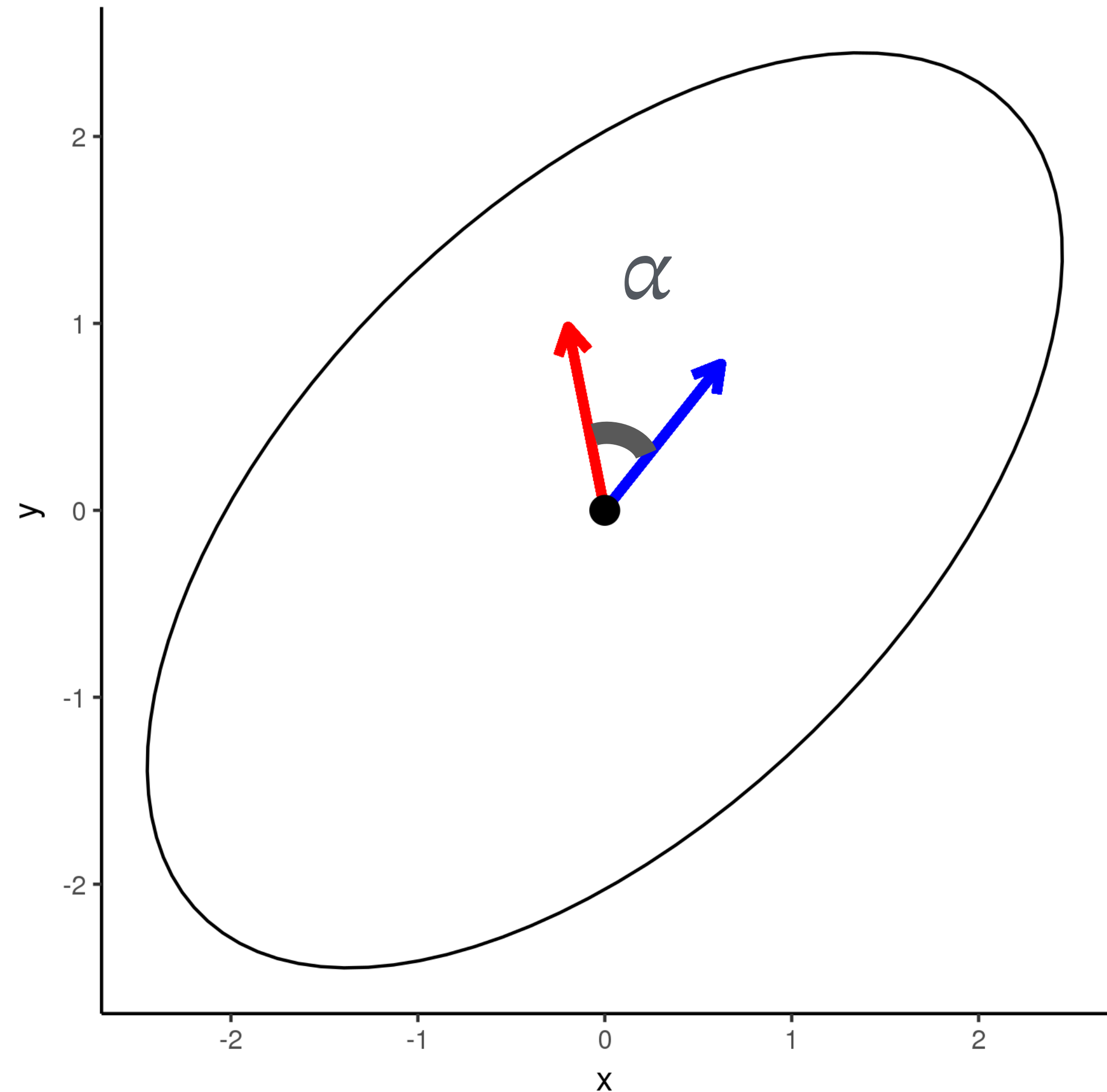
Projeções e variação direcional

$$e(\Sigma, \mathbf{x}) = \frac{\mathbf{x}^T \Sigma \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$$



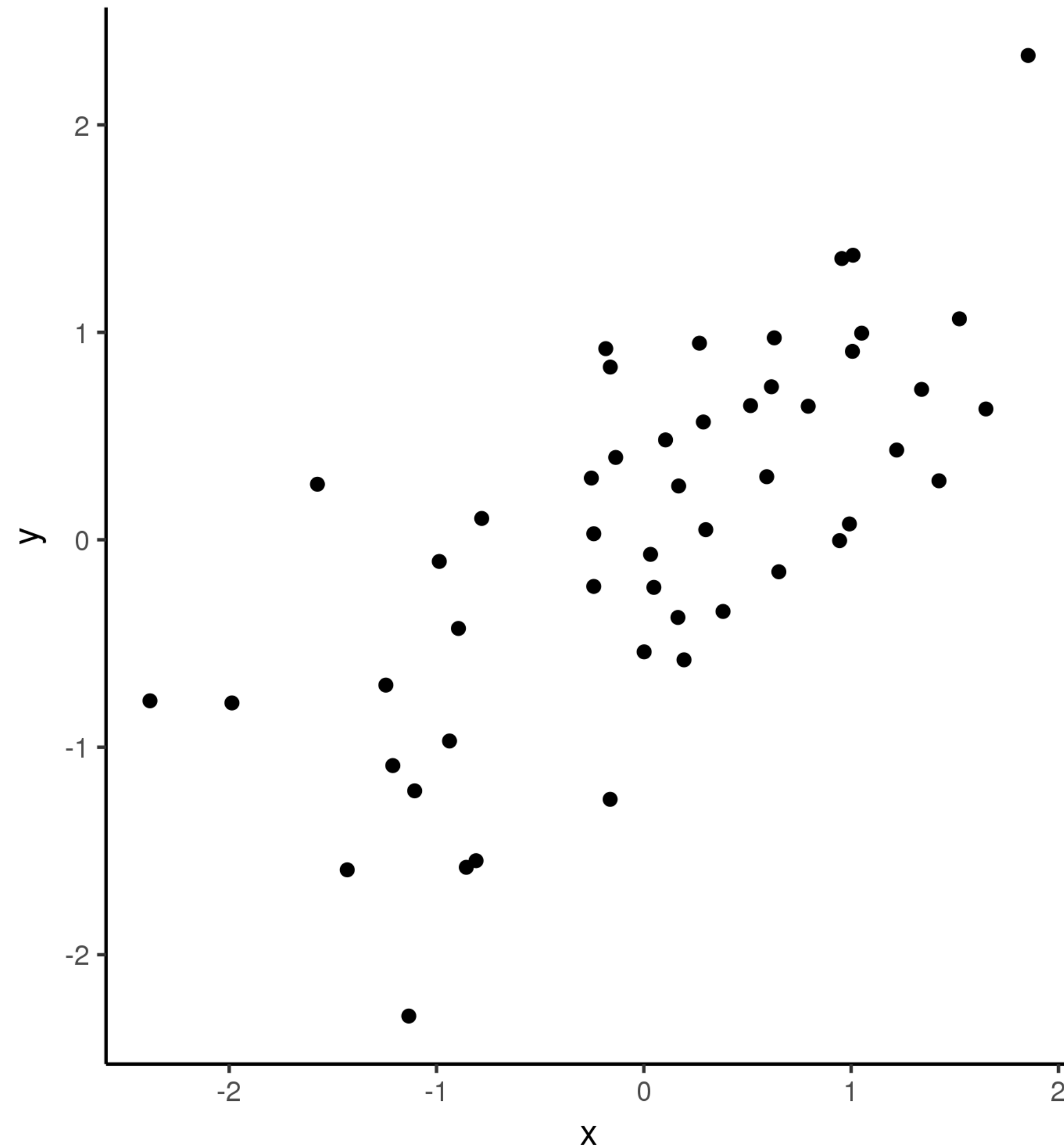
Comparando direções

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\sum_{i=1}^p x_i y_i}{\sqrt{\sum_{i=1}^p x_i^2 \sum_{i=1}^p y_i^2}}$$



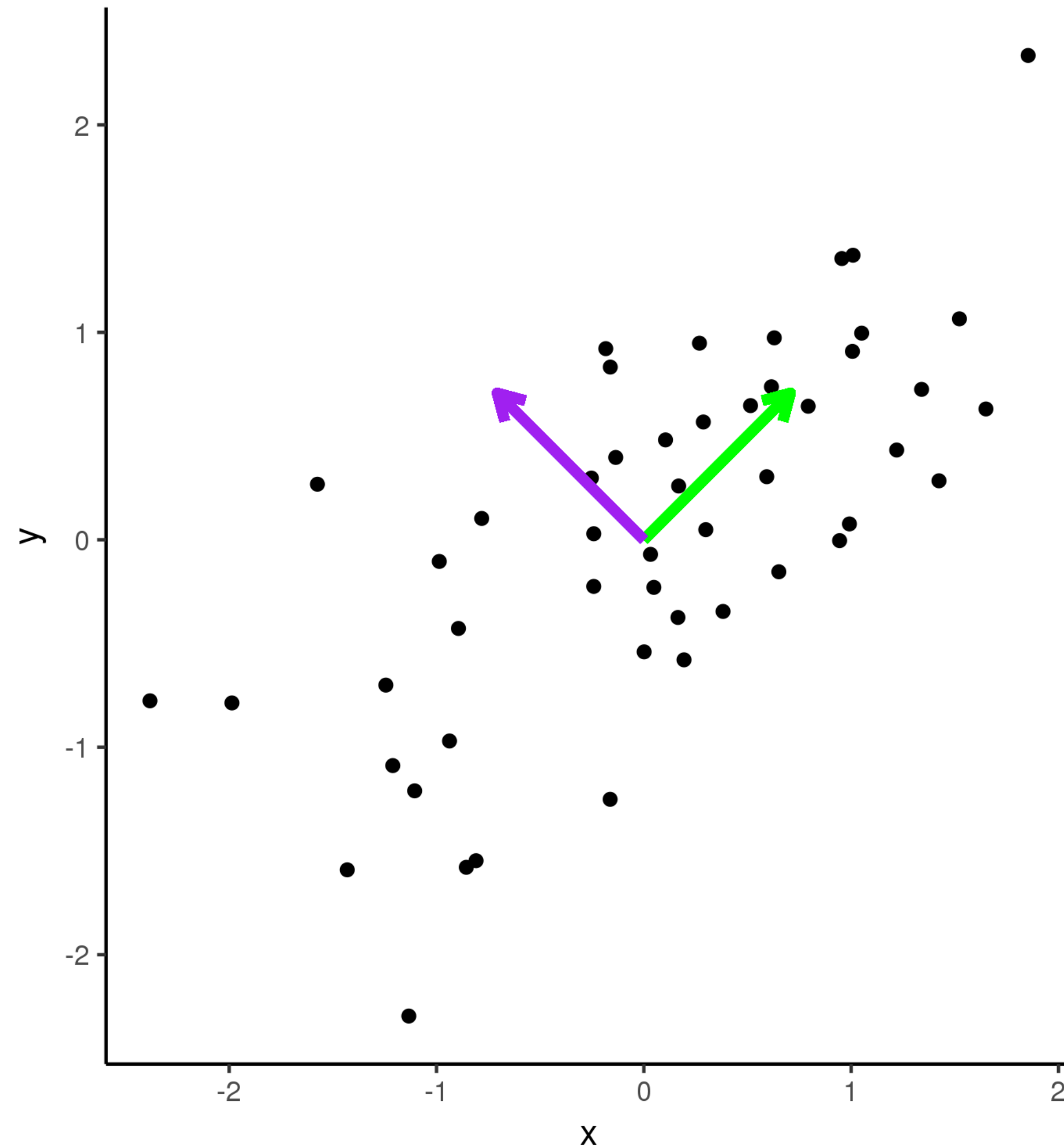
Componentes Principais

- Identificar direções de variação independente



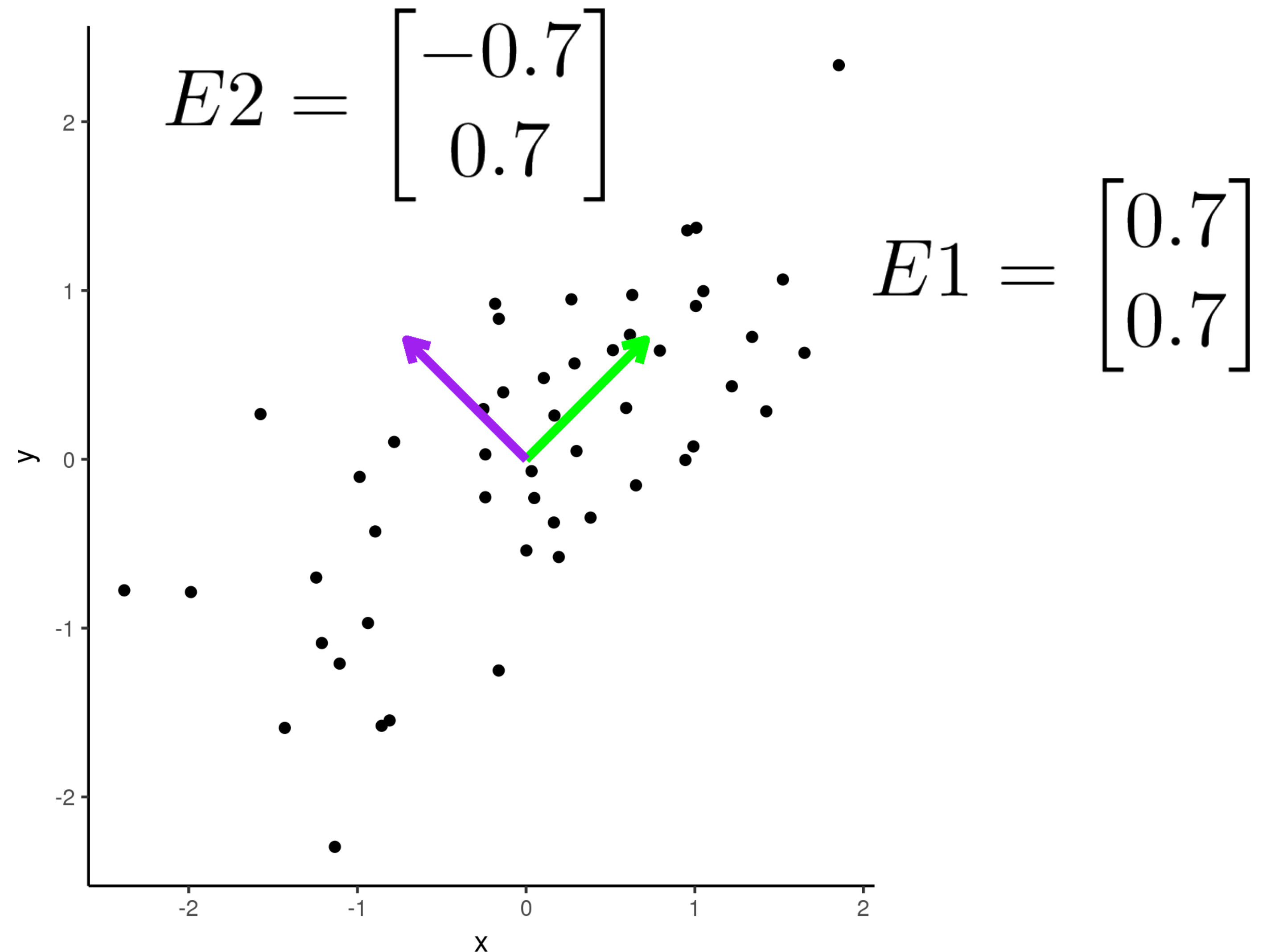
Componentes Principais

- Identificar direções de variação independente

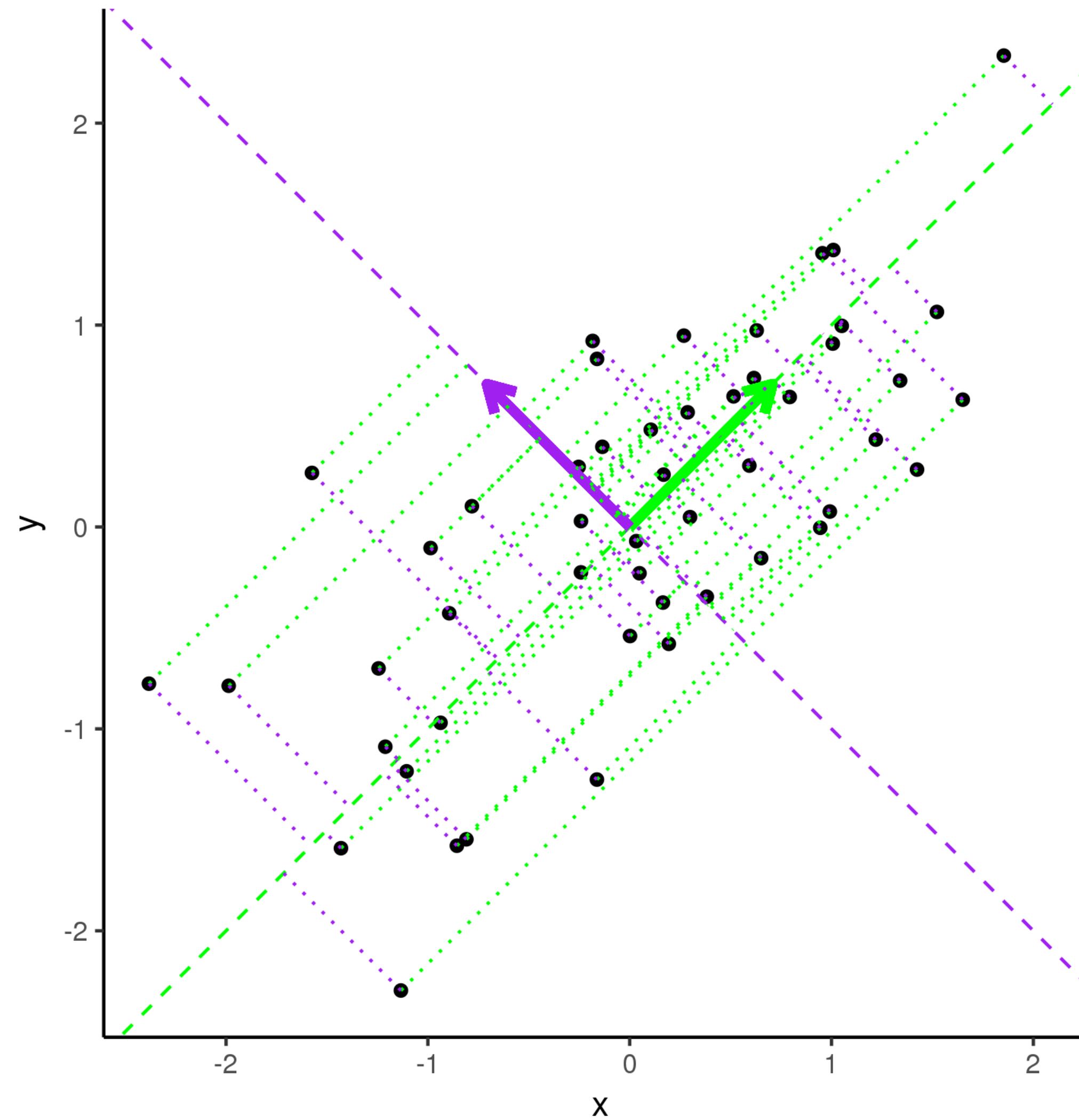


Componentes Principais

- Identificar direções de variação independente
- Direções, ou eixos, definidos são combinações das variáveis originais



Projeções nos PCs



Rotação usando os PCs como eixos

